MBA Thesis

AI Adoption in Reverse Logistics Condition Assessment

An Industrial Economics Perspective on Barriers, Enablers, and Circular Economy Requirements

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Abstract

As telecommunications networks evolve, large volumes of Radio Access Network (RAN) equipment are returned during modernization and vendor-swap cycles and routed through reverse-logistics processes. Determining whether these units should be reused, repaired, or recycled is essential for operational efficiency, cost control, and compliance with circular-economy and sustainability requirements. Today, condition assessment relies heavily on manual inspection, technician judgement, and non-standardized routines, making the process time-consuming, inconsistent, and vulnerable to error. Although Artificial Intelligence (AI) has demonstrated strong potential in inspection and classification tasks, its adoption in telecom reverse logistics remains limited due to variable physical inputs, incomplete documentation, fragmented data sources, and increasing sustainability-driven reporting demands.

This thesis investigates how data quality, workflow characteristics, and sustainability requirements influence the feasibility and organizational readiness for AI-supported condition assessment of returned RAN equipment. The study follows a qualitative single-case design and draws on semi-structured interviews with practitioners across logistics, repair, sustainability, field operations, and AI/ML functions, complemented by internal documents, workflow descriptions, and sustainability reports. This multi-actor perspective provides a 360-degree view of the operational, organizational, and regulatory factors shaping AI adoption in reverse-logistics environments.

Four analytical perspectives guide the analysis. Sociotechnical Systems theory explains the interaction between human routines, tacit knowledge, and technical systems. Data Quality Framework highlights accuracy, completeness, and consistency as prerequisites for reliable AI performance. AI Readiness and Maturity Models assess organizational capabilities, governance, and cross-functional alignment. Circular Economy principles clarify sustainability objectives such as reuse maximization, waste reduction, and regulatory compliance. Together, these frameworks provide an integrated Industrial Economics perspective on AI adoption, linking technology integration to resource efficiency, cost structures, and value recovery.

The findings show that inconsistent defect data, incomplete metadata, and fragmented information systems constrain AI feasibility. Manual inspection routines depend heavily on tacit knowledge, limiting standardization and complicating algorithmic integration. Sustainability objectives create both incentives and constraints, as organizations must meet circular-economy targets while also generating auditable data for regulations such as the EU Corporate Sustainability Reporting Directive (CSRD). Organizational readiness is shaped by coordination practices, knowledge distribution, and governance structures, with significant gaps between strategic ambitions and operational realities.

The study contributes to literature by examining AI adoption in a domain where tacit knowledge, workflow variability, and sustainability pressures intersect, an area largely overlooked in prior research. It offers practical insights for decision-makers seeking to improve reverse-logistics performance, strengthen data governance, and advance circular-economy objectives.

Keywords: AI adoption; reverse logistics; condition assessment; sustainability; circular economy; data quality; telecommunications; industrial economics.

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List of abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ASR	Asset Recovery System
ASP	Approved Service Provider
BBU	Baseband Unit
BSS	Business Support System
CAPEX	Capital Expenditure
CE	Circular Economy
CNN	Convolutional Neural Network
CO_2	Carbon Dioxide
CPE	Customer Premises Equipment
CSRD	Corporate Sustainability Reporting Directive
CU	Customer Unit
DL	Deep Learning
DPP	Digital Product Passport
ECTS	European Credit Transfer and Accumulation System
EoM	End of Maintenance
EoS	End of Support
ESG	Environmental, Social and Governance
EU	European Union
GR	Goods Receipt
LSP	Logistics Service Provider
ML	Machine Learning
NN	Neural Network
NLP	Natural Language Processing
NRO	Network Rollout
OEM	Original Equipment Manufacturer
OSS	Operations Support System
QC	Quality Check
RAN	Radio Access Network
RCO	Return Corrective Order
RNR / R&R	Repair and return
RU / RRU	Radio Unit / Remote Radio Unit
SAP	Systems, Applications and Products
SCM	Supply Chain Management
SKU	Stock Keeping Unit
SLCA	Social Life Cycle Assessment
STS	Sociotechnical Systems
TAT	Turnaround Time
TCO	Total Cost of Ownership

1. Introduction

Telecommunication firms are experiencing changes at a fast pace as the telecommunication and related network technologies continue to advance with time and operators are keen to move toward a more digitized and environmentally friendly infrastructure (Howell & Potgieter, 2023). This development is linked with other trends taking place in industries across the board, such as automation, data-driven decisions, and the use of AI within operations. As the telecommunication enabled networks are getting more complicated and produce more heterogeneous data, telecom providers are increasingly being urged to enhance efficiency, minimize cost of operations, and address concerns pertaining to sustainability through the implementation of practices associated with the circular economy concept.

Within this broader transformation, reverse logistics has become an increasingly important part of the telecom value chain. Large volumes of Radio Access Network (RAN) equipment are returned during upgrade cycles, and determining whether these units should be reused, repaired, or recycled requires careful condition assessment (Cortés Pellicer, Alarcón, & Pérez-Perales, 2025). Unlike other parts of the telecom ecosystem, this assessment process relies heavily on manual inspection, technician experience, and tacit knowledge (Rashid, 2023). The variability of returned equipment, the inconsistency of defect documentation, and the fragmented nature of available data create challenges for standardization and automation. Industry practitioners consistently highlight high 'No Fault Found' (NFF) rates and inconsistent documentation as major challenges that undermine the accuracy and efficiency of condition assessment. At the same time, sustainability-driven objectives, such as waste reduction, reuse maximization, and compliance with circular-economy regulations, place additional demands on the accuracy and efficiency of assessment routines.

While there have been successful implementations of AI-based approaches to perform inspection and classification duties in other fields, the viability of using the same approach in reverse logistics is unknown (Bhowmik, et al., 2024) The presence of different types of inputs, lack of metadata, flexible workflows, as well as sustainability reporting considerations calls into question the ability of AI-assisted evaluation techniques to become embedded within existing business practices. Considering the above challenges, a more comprehensive analysis of sociotechnical, data-related, and organizational factors influencing the adoption of AI technologies is needed.

Below follow the sections detailing the problem, research purpose, and delimitations of the study.

The following sections present the problem discussion, research purpose, and delimitations of the study. Together, they establish the foundation for examining how data quality, workflow characteristics, and sustainability-driven requirements shape the feasibility of AI-based condition assessment in telecom reverse logistics.

1.1. Problem Discussion

AI is increasingly viewed as a tool for improving efficiency, reducing operational costs, and supporting data driven decision making. However, there are certain prerequisites which must be satisfied for the implementation of AI to be successful. These include availability of high-quality data, use of standardized processes, and company readiness to adopt AI. Without these, the application of AI is much harder.

For example, reverse logistics procedures in the telecommunications industry show this difference quite clearly. The returned RAN devices come in various states, ages, and configurations. Information about them is usually incomplete or not always consistent. Defects, moreover, may be described using open text fields, depending on the discretion of a repairman who inspects the device.

Inspection procedure itself relies heavily on tacit knowledge, which makes automation hard to achieve.

On the other hand, the importance of objectives related to sustainability and circular economy principles is gaining momentum. The telecommunication industry's stakeholders must minimize wastage, increase lifecycle of products, and maximize material recovery efforts. To ensure success in such tasks, the need for reliable and effective assessment is critical. Although AI-assisted tools can be used to facilitate these processes, their application is contingent upon solving the issues outlined above.

These problems suggest a definite research gap in the topic as the possibility of implementing AI solutions for reverse logistics assessment remains unclear.

1.2. Problem Formulation and Purpose

This thesis investigates the following research question:

How do data quality characteristics, operational workflow conditions, and sustainability-driven requirements influence the adoption of AI-based inspection methods in the condition assessment of returned telecommunications equipment?

The purpose of the study is to explore the operational, organizational, data-related, and sustainability-driven factors that shape the feasibility of AI-supported condition assessment in telecom reverse logistics. Rather than evaluating specific AI technologies, the study aims to understand how existing workflows, data environments, cross-functional interactions, and circular-economy objectives influence organizational readiness and strategic incentives for AI adoption.

1.3. Delimitations

The scope of this thesis is limited to the condition-assessment process for returned Telecommunication RAN equipment within major telecommunication equipment manufacturer vendors operating in the Nordic region. The study does not evaluate the technical performance of specific AI models or propose new algorithms. Instead, it focuses on the sociotechnical, organizational, and sustainability-related conditions that influence the feasibility of AI-supported condition assessment.

Although the empirical setting is Nordic, the region is home to major telecom equipment manufacturers whose products and reverse-logistics processes are deployed globally. These vendors supply standardized RAN hardware to operators globally, meaning that many of the workflows, defect taxonomies, and sustainability requirements examined in this study are not unique to the Nordic context. As a result, several of the challenges and enablers identified here are likely to be relevant for telecom organizations in other regions as well.

At the same time, the findings should be interpreted with awareness of contextual differences. Regional variations in regulation, documentation practices, warranty rules, and levels of AI maturity may influence how the identified factors manifest in other markets. Therefore, while the study offers insights that are transferable to similar operational environments worldwide, organizations in other regions may need to consider additional local conditions when evaluating AI adoption in reverse-logistics processes.

1.4. Thesis Structure

This thesis follows the structure of the BTH template and is organized into seven chapters. Apart from the introduction chapter 1, the thesis has **Chapter 2** which presents the purpose of the study and formulates the research questions. **Chapter 3** reviews previous research relevant to AI adoption, reverse logistics, and condition-assessment practices. **Chapter 4** outlines the theoretical frameworks that guide the analysis. **Chapter 5** describes the research methodology, including data collection and analytical procedures. **Chapter 6** presents the empirical analysis based on the interview material and supporting documents. **Chapter 7** discusses the findings in relation to the theoretical frameworks and concludes the thesis.

2. Purpose and research question(s)

2.1. Research Purpose

Purpose: The aim of this research thesis is to examine the effect of factors associated with data quality, workflow processes, and sustainability considerations on the possibility of applying inspection technologies based on Artificial Intelligence (AI) in the process of condition assessments of returned Telecommunication equipment specifically Radio Access Network (RAN) hardware, within reverse-logistics operations.

Motivation for the purpose: Prior research on Artificial Intelligence (AI) adoption in manufacturing, predictive maintenance, and network operations highlights recurring challenges related to data quality, workflow standardization, and organizational readiness. However, there is limited empirical work on how these factors play out in the condition assessment of returned units within reverse logistics, particularly in the telecom sector (Bhowmik, et al., 2024). Reverse logistics in Telecommunication equipment hardware represents a distinct context, characterized by heterogeneous Radio Access Network (RAN) configurations, varying documentation practices, and increasing demands for circular-economy and sustainability reporting. These insights from earlier studies informed the choice to focus on data, workflow, and sustainability as key analytical dimensions, while the empirical analysis remains open to additional themes emerging from the interview material.

This research employs an Industrial Economics framework, where the Artificial Intelligence (AI) implementation is not considered only from a technological standpoint but rather as a choice on how to take relevant decisions for resources allocate. In this context, the feasibility of AI implementation can be defined through the cost associated with better governance of data, the economic benefits of having better classification, the costs needed for standardizing work processes, and the added regulatory value related to sustainability tracking.

2.1.1. The Industrial Economics Lens

Industrial Economics studies the allocation of resources, decision-making processes, and generation of value in the face of risk, incomplete information, and conflicting interests (Tirole, 1988). When applied to adoption of technology, the above perspective analyzes investment opportunities based on the cost-benefit structure, competitive dynamics, and efficiencies arising from organizational designs. When applied to adoption of AI technologies for reverse logistics, there are three important concepts within the scope of Industrial Economics.

The first concept is transaction cost economics (Williamson, 1985), which helps analyze the problem of whether to build up one's own AI capabilities in-house, buy off-the-shelf solutions, or simply outsource condition assessment altogether. This decision hinges on the asset specificity and governance costs. Since the AI in question uses the company's proprietary information (data on defects and know-how about the workflow) and is therefore rather specific to the organization, internalization will be the more efficient path.

The second approach, the resource-based view (Barney, 1991), considers data, process expertise, and technical know-how within the company as valuable resources. From this perspective, improvement of data quality becomes an investment in a proprietary resource. This framing helps explain why data-governance investment can generate sustained competitive advantage, even in a cost-center function like reverse logistics.

Thirdly, value chain analysis (Porter, 1985) can be used to situate the condition assessment within the larger context of the value chain of the telecom aftermarket industry. With this framework, it will become evident how artificial intelligence (AI) will change the allocation of costs and value among the different functions in the value chain. The condition assessment process can be considered a cost-center function; the AI-enabled condition assessment can be recharacterized as a value generator through higher reuse, lower scrap, and better sustainability reporting.

In summary, the IE concepts above offer a set of analytical tools for assessing the feasibility of adopting AI that complements the sociotechnical and sustainability frameworks introduced in Chapter 4. While the former frameworks offer insight into why adoption is hard, the latter offers insights into when adoption is economically viable.

2.2. Main Research Question

Guiding the thesis statement is the main research question formulated as follows:

How do data quality, operational workflow conditions, and sustainability requirements influence the adoption of Artificial Intelligence (AI) based inspection techniques in condition assessment of returned telecommunication equipment?

The question is purposefully designed to have three analytical dimensions.

The first one relates to the data-quality dimension, asking if there is enough quality in current information environments to train and deploy an AI solution. The second relates to the workflow dimension, asking about how well current inspection procedures fit into the standardization requirements of AI algorithms. The third one relates to the sustainability dimension, focusing on how circular-economy principles and mandatory environmental reporting might influence AI use.

2.3. Sub-Questions

To facilitate analysis and guarantee thorough examination, the primary research question will be broken down into three sub-questions:

SQ1: How do the characteristics of available data influence the feasibility of applying Artificial Intelligence (AI) based condition assessment in telecom reverse logistics?

This sub-question centers around the information infrastructure required to implement AI technologies. It will examine the nature of data collected in reverse logistics activities and their impact on the viability of applying machine learning techniques. This sub-question will also explore the implications of sustainability disclosure requirements for the further improvement of data governance through AI. From an Industrial Economics perspective, this sub-question relates to how information asymmetries, data quality, and transaction costs influence resource allocation decisions in reverse-logistics condition assessment.

SQ2: How do existing inspection routines and workflow conditions shape the possibilities for integrating Artificial Intelligence (AI) based condition assessment?

This sub-question examines the process dimension of AI adoption. It investigates how existing human centered workflows, characterized by sensory judgement, site-specific practices, and undocumented decision logic, interact with the standardization requirements of AI systems. It further considers how circular-economy targets, such as reuse maximization and waste reduction, shape routing decisions and the criteria against which AI outputs would need to be evaluated. In Industrial Economics terms,

this sub-question highlights how workflow design and process standardization affect operational efficiency, coordination costs, and the economic viability of AI adoption.

SQ3: How do organizational structures, coordination practices, and sustainability governance influence readiness for Artificial Intelligence (AI) adoption?

The sub-question considers organizational and institutional issues associated with the adoption of AI technologies. The research will focus on cross-functional coordination, data ownership and competencies, and how organizational governance mechanisms aligned with sustainability objectives relate to operational considerations. Organizational readiness is not viewed here as a fixed characteristic but rather as a dynamic capacity. From an Industrial Economics viewpoint, this sub-question addresses how governance structures, organizational design, and incentive alignment shape the conditions under which AI investments become feasible and strategically attractive.

Figure 1 illustrates the decomposition of the main research question into its three sub-questions and maps each sub-question to the primary analytical dimension it addresses. The structure itself takes into account the multiple layers involved in the implementation of AI in this setting, whereby the potential for AI implementation is influenced both by the nature of data at hand and by the attributes of the current workflow processes, along with the organizational factors that may facilitate or hinder such an implementation. Each sub-question is analyzed using the appropriate theoretical perspectives discussed in Chapter 4.

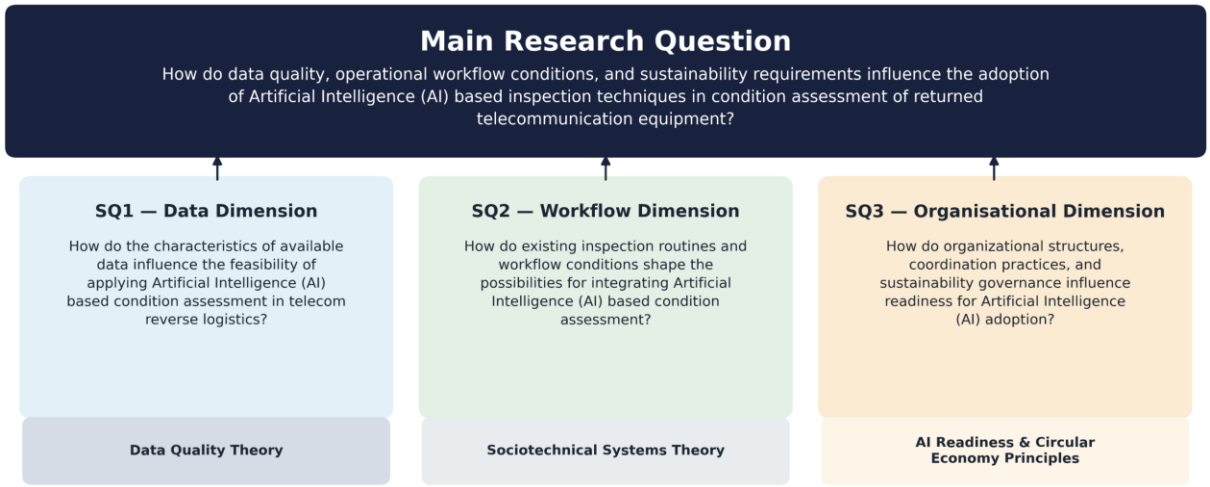


Figure 1 Decomposition of the main research question into three sub-questions.

2.4. Research Contribution

The study makes three interconnected contributions: theoretical, empirical, and practical. Theoretically, it applies and integrates four complementary frameworks which are listed here as Sociotechnical Systems (STS) theory, Data Quality Framework, AI Readiness and Maturity Models, and Circular Economy principles - into a single analytical architecture. Each framework has been applied individually in related domains, their combined use to examine AI adoption in reverse-logistics condition assessment has not been explored in prior research. The empirical contribution of this work is the provision of qualitative insights on how the major Nordic telecom equipment manufacturers operate a reality that is not well-captured in literature currently available. Practically, the study offers a structured way for decision-makers in aftermarket operations, supply chain management, and sustainability to evaluate organizational preparedness for AI-supported condition assessment.

Figure 2 illustrates how the study's scope fits within the broader telecommunications value chain, distinguishing between activities included in the analysis and those outside its boundaries. The focus is limited to the assessment of returned RAN equipment (basebands, radios, and antennas) within telecommunication equipment manufacturers in Nordic region. The study does not evaluate specific AI algorithms or their performance in downstream network operations.

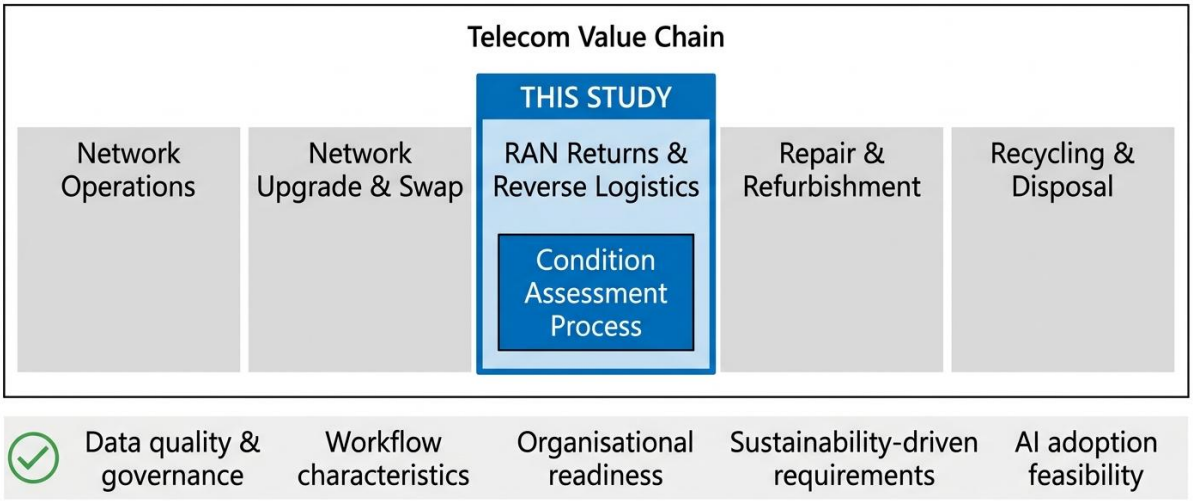


Figure 2 Research scope and delimitation boundary.

2.4.1. Theoretical contribution

Theoretically, the study integrates four frameworks, Sociotechnical Systems theory, Data Quality Framework, AI Readiness and Maturity Models, and Circular Economy principles, into a single analytical architecture applied to a setting that no prior study has examined through this combined lens. Individually, each framework has been applied to related domains: STS theory to manufacturing automation, Data Quality Framework to business intelligence systems, AI Readiness models to R&D environments, and Circular Economy principles to sustainability management. What is new is their joint application to reverse-logistics condition assessment, where the interaction between these dimensions is not additive but systemic: data problems are sociotechnical in origin, readiness constraints are exacerbated by sustainability demands, and workflow variability undermines both data quality and organizational learning. The integrated framework developed here provides a replicable analytical template for future studies of AI adoption in similarly complex, physically heterogeneous, and sustainability-sensitive operational settings.

2.4.2. Empirical contribution

From an empirical perspective, the study presents in-depth qualitative information from one of the largest Nordic telecom vendors active in various regions and functional disciplines. The interview material contains practical insights into the realities of NFF rates, informal inspections, data limitations from customers, sustainability reporting pressure, all of which are completely ignored by the scholarly literature on AI integration, focusing only on R&D and networking operations involving data that is already formatted. Through such detailed observations, the study adds a set of empirical benchmarks that professionals and academics can rely on for understanding how much time and money would be needed for integrating AI technologies in addition to the prerequisites that need to be fulfilled before implementation. It should also be noted that conducting eleven interviews in different areas, including logistics, repairs, sustainability, field operations, and academic knowledge,

was enough to achieve thematic saturation, which constitutes a useful insight for future qualitative studies on the topic.

2.4.3. Practical contribution

In practice, the proposed methodology provides a well-structured diagnostic approach based on the four analytical perspectives and tested by empirical evidence that decision makers in after-market operations, supply chain management, and sustainability can utilize to evaluate their organization's preparedness to implement AI for condition assessment. Instead of suggesting an optimal AI tool, the proposed method defines a chronological order of necessary steps (investments, data management prior to machine learning modeling, standardization of workflows prior to hybrid integration, alignment of sustainability prior to implementation), which must be undertaken sequentially. This logical ordering is not found in any current AI adoption guides produced by technology vendors, who mainly concentrate on technological solutions, whereas organizational prerequisites have been largely overlooked. The proposed contribution in its practical application is, therefore, most applicable to the operations director, head of digital transformation, and sustainability manager in telecom companies dealing with similar problems.

Having established the purpose, research questions, and expected contributions of this study, the following chapter situates the research within the existing body of literature. Chapter 3 reviews empirical and theoretical work on AI adoption in telecommunications and supply chain contexts, examines prior studies on organizational and sociotechnical barriers to automation, and identifies the specific research gaps that motivate the present investigation. This review forms the foundation upon which the theoretical frameworks introduced in Chapter 4 are selected and justified.

3. Previous research

Research on artificial intelligence (AI) in telecommunications has grown rapidly in recent years, but most studies focus on network optimization, predictive maintenance, and automation in R&D or operations. It is less common for researchers to investigate how firms use artificial intelligence in their return logistics process, where returned telecom RAN equipment must be inspected, classified, and routed under conditions that are often less structured than those found in manufacturing or network operations. Prior research suggests that reverse-logistics environments may involve heterogeneous equipment, variable documentation practices, and fragmented information systems, but these tendencies have not been systematically investigated in the telecom sector.

Many firms tend to prioritize profit-generating functions, cost-center activities such as repairs, returns, and sustainability reporting receive less attention in both practice and academic research. As a result, the feasibility of Artificial Intelligence (AI) systems in operationally complex, data-limited environments remains insufficiently understood. This chapter reviews empirical and theoretical studies relevant to the present investigation, identifies the gaps that motivate the research, and explains how prior literature informed but did not predetermine the analytical dimensions used in the research questions.

3.1. Empirical and Theoretical Studies on AI Adoption

The studies reviewed in this chapter form the empirical and theoretical foundation for the present research. They address themes central to the thesis: organizational readiness, human and AI interaction, data quality in operational settings, and AI feasibility in logistics and supply chain functions. The review begins with qualitative studies conducted in Nordic telecommunications environments, followed by cross-industry research on AI readiness, theoretical work on automation–augmentation dynamics, and systematic reviews of AI in supply chains. Together, these studies provide sensitizing concepts that informed the development of the research questions while leaving the empirical analysis open to new themes emerging from the interviews.

3.1.1. AI adoption barriers in telecommunications

The research of Verma (2023) and Dafallah (2023) provides contextually relevant insights. Verma conducted a study of AI implementation in Swedish and Finnish telecom research and development departments which revealed five interconnected obstacles: non-specialist staff members who lack AI understanding, difficulties in finding qualified personnel, data protection issues, excessive infrastructure expenses, and the existence of separate departmental data systems. Dafallah researched Ericsson's research and development department to show how data problems and simulation expenses and knowledge gaps and management issues and teamwork difficulties and employee safety worry all work together to stop artificial intelligence from being used in their production systems. The two studies suggest recurring tendencies in telecommunications operational environments which include data fragmentation and inconsistent operational processes and lack of cross-departmental operational oversight. These studies indicate that such tendencies may become more visible in departments which have lower digital capabilities than in research and development settings. The research of (Davenport, 2018) presents an industry-wide comparison which shows that both organizational and data-related obstacles prevent companies from implementing AI technologies more than technical challenges do. The research shows that the most effective AI value creation occurs through process automation and cognitive insight functions which use structured operational data and high-quality data. These findings are relevant for understanding the conditions under which AI-supported condition assessment may be feasible.

3.1.2. Organizational readiness and human–AI collaboration

The research by Jöhnk, Weißert, and Wyrtki (2021) examined operational environments through interviews with 27 participants from various industries to create an AI readiness model that shows organizations employ incorrect evaluation methods because they consider their technological capabilities while they fail to recognize the essential requirements of process standardization and data governance. The research found that resistance to change from cultural factors which especially impact teams with deep knowledge of their work processes proved to be the biggest challenge which people failed to recognize. The automation-augmentation paradox which Raisch and Krakowski (2021) studied shows that AI systems achieve optimal performance when they handle tasks with predictable outcomes while standardized data, but organizations need augmentation when their input data requires human assessment in situations with serious outcomes. Their complementarity concept informs the analytical lens used later in the thesis, which shows that AI handles structured volume processing while humans maintain control over uncertain situations that have serious consequences. Jarrahi (2018) proves this argument through a human-AI symbiosis model which demonstrates that trust and transparency together with human override systems create the conditions for achieving complementarity, yet these requirements are often less established in reverse-logistics environments. Chowdhury, Budhwar, Dey, Joel-Edgar, and Abadie (2022) present a vital empirical result which shows that AI adoption in knowledge-based operational environments stops because practitioners hold decision-critical information in their mental frameworks and organizations need to invest continuously before they can implement AI training at development stage.

3.1.3. Data quality and supply chain AI

The study conducted by Toorajipour et al. (2021) examined 98 peer-reviewed studies about AI applications in supply chain management which showed that organizations use AI technology for inspection and returns management processes less often than they use it for inventory management and forecasting because of three main obstacles which are data inconsistency and missing domain-specific labeled datasets and difficulties to connect legacy systems. Their review discovered a research gap at the sustainability AI intersection because there are no studies that examine how circular-economy targets and environmental compliance requirements shape AI adoption which the present study addresses directly. Budach et al. (2022) establish that model performance depends on data quality which exhibits a non-linear relationship because performance drops suddenly when data quality falls below specific thresholds. The authors suggest that organizations may need to reach certain data-quality thresholds before AI systems can deliver meaningful benefits from their AI systems because incremental governance enhancements below threshold levels deliver no apparent AI advantages according to the non-linear relationship.

3.1.4. Reverse logistics, inspection, and sustainability

The research by Hangl (2022) shows that industrial AI implementation faces multiple obstacles because organizations struggle with data sharing between departments while their employees lack necessary skills and their employees refuse to accept new practices and their leaders operate without established rules, and their operations receive restricted managerial backing. Bhowmik et al. (2024) demonstrate that AI-supported inspection performs reliably in manufacturing settings where high-quality labelled datasets and uniform physical inputs are available, but the authors note that these conditions are often less common in reverse-logistics environments because input variability creates different requirements which need specific solutions. Cortés Pellicer, Alarcón, and Pérez-Perales (2025) show that increasing sustainability requirements now transform operational assessment into a strategic assessment that directly affects corporate sustainability results which organizations need to assess their AI investments at the same time bring new requirements for data tracking which their existing systems do not support.

3.2. AI in Reverse Logistics and Inspection Processes

Research specifically addressing AI in reverse logistics is comparatively limited. Studies in adjacent fields, such as manufacturing quality control and automated visual inspection, demonstrate that AI can effectively support defect detection when high-quality labelled datasets are available and physical inputs are sufficiently uniform (Hangl, 2022; Bhowmik et al., 2024). The success of these systems relies on prerequisites which reverse-logistics environments do not typically maintain. Returned equipment shows substantial variability because it consists of different types of equipment that have been used in different ways and maintained in different ways since their initial installation. Prior studies suggest that defect data may be incomplete, inconsistently recorded, or described through free-text technician notes rather than structured taxonomies. Legacy information systems may limit integration with contemporary analytics tools, resulting in data-access constraints. Existing AI inspection models may require adaptation before they can be applied because of these characteristics which require researchers to develop specific solutions for the data and operational challenges that exist in reverse logistics. The absence of such research constitutes one of the primary gaps motivating the present study.

AI systems show different performance levels because manufacturing inspection requires different capabilities than reverse logistics assessment needs. Manufacturing inspection requires new assets which maintain their exact dimensions and undergo controlled lighting tests while quality engineers use standard procedures to create their equipment identification system. Reverse logistics manages returned items that have been used in different field situations and have potential customer-driven changes which need to be handled with insufficient documentation and variable inspection methods by technicians who have different levels of expertise. The authors Cortés Pellicer Alarcón and Pérez-Perales 2025 demonstrate that this distinction needs a new understanding of AI inspection systems when they want to operate in reverse logistics environments. Organizations need to create data governance systems and workflow procedures which generate operational conditions that existing manufacturing facilities already possess instead of using standard manufacturing AI systems.

3.3. Organizational and Sociotechnical Barriers

Multiple research studies demonstrate that organizational elements function as essential components which organizations frequently disregard during their process of implementing artificial intelligence technology. Hangl (2022) identifies a recurring cluster of barriers across industrial AI implementations: lack of cross-functional coordination leads to fragmented data ownership and misaligned incentives; insufficient digital skills leave frontline staff ill-equipped to interpret or trust AI outputs; resistance to change, whether driven by habit, professional identity, or job-security concerns, slows the adoption of new workflows; unclear governance structures create ambiguity about who is responsible for AI systems when they produce errors or require updating; and limited managerial support deprives AI initiatives of the resources, sponsorship, and organizational legitimacy needed to achieve sustainable integration. The reverse logistics field requires these findings because condition assessment needs three different functions, which are logistics and repair and sustainability and quality, to work together because each function has its own data gathering methods and performance evaluation criteria. AI adoption needs to complete its process because the sociotechnical system requires all groups to collaborate while technical implementation only delivers partial results. Organizations need to establish governance frameworks together with change management processes while they develop mutual agreements about operational usage of AI results. The implementation of AI systems in organizational environments faces difficulties because organizations allocate duties throughout their reverse-logistics activities. Assessment of conditions requires input from multiple professionals including logistics staff members, repair technicians, sustainability coordinators, and data engineers who work in different locations and time zones and across multiple organizational levels. Jarrahi (2018) states that successful human–AI cooperation needs both parties to share identical mental frameworks about AI capabilities and limitations which

becomes challenging when team members work from different locations and have not yet used AI technologies. The shared understanding deficiency leads to two problems which include hindering adoption progress and establishing conditions that lead to passive resistance when users disregard AI results without direct confrontation thus creating a challenging situation for solution identification and resolution.

The third dimension of the knowledge-capture problem exists as a separate dimension. Research in STS shows that tacit expertise which combines sensory judgement knowledge and experienced knowledge with situation-specific pattern recognition remains impossible to formalize, which makes its conversion into AI training data unachievable. Technicians often describe relying on sensory cues such as listening to connector sounds, touching surface materials, and drawing on past experience with similar units. The system lacks any data about these cues, which makes it impossible to create labels for them after the fact. Chowdhury, Budhwar, Dey, Joel-Edgar, and Abadie (2022) show that AI adoption in knowledge-intensive operational settings stalls not because of a lack of data volume but because the most decision-relevant information exists only in practitioners' heads. Prior research suggests that knowledge externalization may require sustained investment and structured methods to capture technician knowledge because this process both precedes and needs to happen before AI training starts, yet this step remains absent from current AI implementation plans used in the telecom industry.

3.4. Sustainability and Circular-Economy Considerations

The recent studies demonstrate that sustainability has become an essential factor which businesses use for making operational choices in all industrial sectors, including telecommunications. The circular-economy goals require organizations to achieve maximum product reuse while decreasing their waste output and enhancing their ability to recover materials throughout the entire product lifecycle (Geissdoerfer, 2017; Ellen MacArthur Foundation, 2013). Telecom reverse-logistics processes are increasingly influenced by these goals because the condition-assessment process needs accurate unit classification to determine the best recovery method, which includes equipment reuse and refurbishment and recycling. Organizations may need to explore AI-supported inspection methods because these objectives drive better consistency and classification accuracy. The sustainability requirements demand organizations to collect new data while they report their findings about material tracking and carbon reduction results and their compliance with regulations like the EU Corporate Sustainability Reporting Directive (CSRD), which creates difficulties for AI implementation because existing systems cannot produce the necessary records.

The regulatory dimension of sustainability is an increasingly important structural driver of operational change. The EU Corporate Sustainability Reporting Directive (CSRD), which entered into force in 2024, requires large organizations to disclose detailed information on material flows, waste reduction, and carbon emissions across their operations. Telecom vendors must ensure that all reverse-logistics data which includes their condition-assessment outcomes and routing decisions and scrap rates becomes traceable and auditable and accessible for external verification purposes. This requirement creates an indirect yet powerful push which drives organizations to enhance their data governance practices within the returns process because sustainability reporting needs structured consistent records which serve as the foundation for AI-supported assessment. Cortés Pellicer, Alarcón, and Pérez-Perales (2025) observe that this regulatory convergence is beginning to reframe condition assessment from a purely operational function into a strategic capability with direct implications for corporate sustainability performance, a reframing that considerably strengthens the business case for AI investment.

The academic literature shows a noticeable gap between artificial intelligence and circular economy requirements despite the existing incentives. Toorajipour et al. (2021) conducted a systematic review

of 98 studies which showed limited research on how sustainability reporting requirements and reuse targets and environmental regulations impact AI implementation. This gap has been noted in prior research. The research establishes sustainability as an essential element which determines AI implementation capacity through its analysis of how circular-economy goals create both financial incentives and operational challenges for AI technology implementation. Prior studies indicate that sustainability can act as both a driver and a constraint for adoption of AI.

3.5. Summary of Research Gaps

The current investigation requires multiple interrelated research gaps which the literature review shows to exist. Empirical research on AI adoption in telecom reverse logistics remains limited, leaving practitioners with little operational decision-making evidence. The specific data-quality challenges which arise in returned-equipment environments need better understanding because AI adoption literature assumes structured data contexts. The current research setting needs to study how manual inspection routines which depend on tacit knowledge interact with AI systems which need codified structured inputs because this relationship determines integration possibilities. AI adoption frameworks do not incorporate sustainability-based requirements because they treat operational and regulatory objectives as separate elements, instead of showing how these objectives relate to each other. The field of AI implementation for reverse logistics within organizations still needs research because previous studies have focused on R&D and network operations. These gaps motivate the present study and define its analytical scope because they define their analytical scope and require a qualitative exploratory method to reveal complex elements of this poorly researched situation.

Table 1 maps the above reviewed studies against the five research gaps identified above, illustrating which combinations of context, data, sustainability, and organizational dimensions each study addresses and where the present thesis contributes new coverage.

Table 1 Coverage of key research dimensions across reviewed studies.

Study	**Reverse Logistics Context**	**Data Quality Challenges**	**Tacit Knowledge & Workflow**	**Sustainability–AI Intersection**	**Organizational Readiness**
Verma (2023)	✗	Partial	✗	✗	✓
Dafallah (2023)	✗	✓	✗	✗	✓
Davenport & Ronanki (2018)	✗	Partial	✗	✗	✓

Study	Reverse Logistics Context	Data Quality Challenges	Tacit Knowledge & Workflow	Sustainability–AI Intersection	Organizational Readiness
Jöhnk et al. (2021)	✗	Partial	✗	✗	✓
Raisch & Krakowski (2021)	✗	✗	✓	✗	Partial
Jarrahi (2018)	✗	✗	✓	✗	Partial
Chowdhury et al. (2022)	✗	Partial	✓	✗	Partial
Toorajipour et al. (2021)	Partial	✓	Partial	Partial	✓
Budach et al. (2022)	✗	✓	✗	✗	✗
Hangl (2022)	Partial	Partial	Partial	✗	✓
Bhowmik et al. (2024)	✓	✓	Partial	✗	✗
Cortés Pellicer et al. (2025)	✓	Partial	Partial	✓	Partial
This study	✓	✓	✓	✓	✓

Gap 1: Limited research on AI in telecom reverse logistics Previous studies have examined AI adoption in telecom R&D, network operations, and manufacturing inspection, but none have focused specifically on reverse-logistics condition assessment. While Bhowmik et al. (2024) and Hangl (2022) discuss inspection and industrial AI challenges, they do not address the variability,

documentation practices, or operational constraints unique to returned RAN equipment. This study fills the gap by providing empirical insights from practitioners directly involved in telecom reverse-logistics processes.

Gap 2: Insufficient understanding of data-quality challenges in returned-equipment environments Several studies highlight data-quality issues in supply chains (Toorajipour et al., 2021; Budach et al., 2022), but they focus on structured datasets or manufacturing contexts. They do not examine inconsistent defect descriptions, missing metadata, or fragmented install-base information typical of telecom returns. This study contributes by showing how these specific data-quality limitations shape AI feasibility in condition assessment.

Gap 3: Lack of research on tacit knowledge and workflow variability in condition assessment Research on human–AI collaboration (Raisch & Krakowski, 2021; Jarrahi, 2018) emphasizes the importance of tacit knowledge but does not apply these insights to reverse logistics. Existing telecom studies focus on R&D or network operations, where workflows are more standardized. This study addresses the gap by examining how sensory judgement, undocumented decision logic, and site-specific routines influence AI integration in condition assessment.

Gap 4: Minimal integration of sustainability requirements into AI adoption frameworks Although circular-economy principles and sustainability reporting are widely discussed (Geissdoerfer, 2017; Cortés Pellicer et al., 2025), prior research rarely connects these requirements to AI adoption. Toorajipour et al. (2021) identify this as an underexplored intersection. This study responds by analyzing how CO_2 targets, reuse goals, and CSRD-driven reporting obligations shape both the incentives and constraints for AI-supported inspection.

Gap 5: Limited examination of organizational readiness in operational, non-R&D contexts AI readiness studies (Jöhnk et al., 2021; Davenport & Ronanki, 2018) focus on strategic or R&D environments, not on operational areas such as reverse logistics. They do not consider fragmented governance, cross-functional coordination, or customer-imposed data restrictions typical in returns processes. This study fills the gap by evaluating organizational readiness across logistics, repair, sustainability, and field operations.

3.6. Empirical Reinforcement of the Research Gap

The literature shows these problems through theoretical concepts, but the study's empirical results show their actual effects on operations. The interview data illustrates and further contextualizes the patterns identified in prior research, revealing how these issues manifest in the daily routines of practitioners managing RAN returns across multiple regions and functions. Data fragmentation emerged as a pervasive and multi-dimensional challenge. Interviewees confirmed the presence of inconsistent defect descriptions, incomplete metadata, missing install-base information, and unreliable documentation across the systems used to manage returned equipment. The data quality problems which researchers have documented in existing literature appear in these problems which demonstrate how AI models may struggle to perform effectively within reverse logistics settings because their training data lacks sufficient strength and accuracy and complete information. The workflow showed two important aspects because technicians employed their hidden skills and personal judgement while following different work procedures which varied between locations and teams and products. Supplier response times fluctuated without a set pattern because return processes depended on specific contract terms and local practices determined how troubleshooting procedures were implemented at specific sites. The variability between systems matches the sociotechnical challenges which previous studies have discovered, showing that AI implementation requires standardization to enable algorithms to learn different tasks.

The interviews showed that people required unspoken knowledge to perform physical inspections and assess damage and make routing choices. The need for human specialists who make decisions based on their practical knowledge shows that current condition assessment methods get used because they lack sufficient structured guidelines to make automated systems work. Existing studies about AI implementation in telecom do not investigate this gap which exists between human judgment and standard records in condition assessment. Sustainable development practices effectively function as both operational drivers and operational constraints because they create CO_2 reduction targets and scrap reduction targets and component reuse targets which create operational requirements but require additional reporting. The research findings demonstrate that previous studies reached a research gap because they failed to include circular-economy requirements when evaluating AI implementation. The organization faced multiple obstacles because their IT governance system remained underdeveloped and their systems lacked cohesion and their customers restricted AI access and their organization lacked sufficient AI expertise which caused their reverse logistics AI readiness to stay behind it their R&D and network operations AI development. The research gap receives validation because the empirical evidence shows that existing research fails to investigate how data fragmentation and workflow variability and tacit knowledge and sustainability pressures and organizational constraints impact the implementation of AI condition assessment systems in telecom reverse logistics. The interviews show that these factors are experienced as part of daily operational work, but they exist as fundamental parts of the daily work life for practitioners, highlighting the relevance and timeliness of the present study.

4. Theory

The following chapter discusses the theories employed in examining the potentiality of using artificial intelligence in condition assessment in telecom reverse logistics. While there is extensive literature regarding the use of AI in fields such as network optimization, predictive maintenance, and customer analysis, a unique approach to the implementation of AI in the field of assessment of returned Radio Access Network (RAN) equipment will be required due to the nature of the reverse-logistics environment, which features heterogeneity of inputs, fragmentation of information, tacit knowledge, and the necessity to report sustainably. To address these complexities, the chapter employs five theories, including: Artificial Intelligence as the technological basis; Sociotechnical Systems (STS) theory in order to examine the relationship between technology and human labor; Data Quality Framework to evaluate the compatibility of information contexts with AI algorithms; AI Readiness and Maturity Models to determine organizational readiness for implementing AI; and Circular Economy principles to consider the sustainability-based drivers of AI use.

In this thesis, a distinction is made between formal theories and analytical frameworks or perspectives. Sociotechnical Systems (STS) is treated as a formal organizational theory that offers explanatory propositions about how social and technical subsystems interact. By contrast, the data-quality dimensions proposed by Wang and Strong, AI readiness and maturity models, and circular-economy principles are used as analytical frameworks or perspectives that structure empirical observations rather than as fully developed theories in the strict sense. This clarification is important to avoid overstating the theoretical status of constructs that primarily function as conceptual tools for organizing the analysis.

4.1. Telecom Reverse Logistics Context for AI Adoption

It should be noted that the objective of this section is not to provide definitions related to Telecom terminology. Rather, it aims at outlining the operational features of telecom reverse logistics and thus laying down the context, within which theoretical perspectives will be analyzed below.

The use of artificial intelligence (AI) in telecom reverse logistics should consider these operational features of returned Telecom equipment rather than provide detailed descriptions of returned equipment. Age, Hardware configuration, Software status, and Condition of returned equipment are highly variable, and documentation provided together with the returned equipment is often incomplete or inconsistent and therefore these variables generate unpredictable input data. As AI-based assessment systems require stable data structures, labelling methods, and procedures used.

Condition assessment requires physical inspection of returned units and analysis of their operation and maintenance history, however, neither process is performed according to unified standards. Lack of uniformity in dismantling procedure, inconsistent defect classification language, and absence of a comprehensive tracing of equipment serial numbers make training datasets impossible to assemble. Consequently, telecom reverse logistics cannot be compared to manufacturing inspection or network operations monitoring, which generate structured data.

4.1.1. Artificial Intelligence and Its Subfields

The relevance of artificial intelligence (AI) in this study lies not in the mechanics of machine learning algorithms but in the alignment between algorithmic requirements and the operational characteristics of telecom reverse logistics. AI-based inspection relies heavily on structured and representative data.

In case of Telecom, the equipment such as RAN equipment returns, defect descriptions may differ from one technician to another, capturing of images may be inconsistent, while additional metadata may not always be provided. As a result, it becomes impossible to implement any form of supervised learning even for more sophisticated models.

Deep learning methods can support visual assessment when imaging conditions and defect taxonomies are standardized, however, the standardized process is rarely seen when it comes to dismantling and reverse flows. Text analysis algorithms may help structure data from technician's records, but this is possible only if the terminology used is standardized and descriptive enough. Failure to meet these requirements means that AI could worsen inconsistencies.

Machine Learning (ML) models can support analyzing condition assessment when defect categories are standardized and historical examples are available in sufficient volume. The system works well when users input data which stays constant throughout the process while designers use labeled information to model actual patterns within the dataset. The telecommunications industry employs ML to detect anomalies, predict traffic patterns, and analyze customer data according to research by (Parada R. M.-S., 2018) and (Soleimani, 2019). The supervised ML method uses labeled training data to develop models which assess conditions by classifying returned units into specific categories for reuse, repair, or recycling. Supervised learning requires access to large datasets that contain precise historical information which makes it challenging to implement reverse logistics operations where labeling practices have not maintained consistency and defect categories have not established standard definitions.

Deep Learning (DL) operates as a machine learning (ML) branch that employs neural networks with multiple layers to handle unstructured data which includes both images and unstructured text describing defects. DL models operate through their ability to automatically identify important features which enable their use in visual inspection tasks that share conceptual links with condition assessment work according to (Parada R. A.-Z.-G., 2021). The **Convolutional Neural Network (CNN)** deep learning architecture achieves excellent results when it performs automated visual defect detection in manufacturing quality control, which involves inspecting high-resolution component images to determine their faulty or acceptable status. RAN equipment inspection can use CNN technology to identify physical damage, which includes corrosion and connector wear and mechanical deformation. The system needs three requirements to function properly: it requires consistent image acquisition conditions and standardized defect categories, and it needs adequate training data, which includes labeled examples. **Natural Language Processing (NLP),** another DL application, could be used to extract structured information from free-text defect descriptions recorded by technicians, converting unstructured narrative entries into categorical features suitable for downstream classification. The underlying text consistency and text informativeness determine the approach feasibility, because these two factors exhibit significant variations among different sites and technicians.

Both ML and DL need high quality datasets which need to be both consistent and sufficiently large. The reverse logistics system needs equipment to be returned but this process fails because returned equipment shows multiple types of issues while defect records display no standard format. The unstructured nature of operational environments makes AI implementation more difficult compared to structured operational fields. Another factor to consider involves understanding how discriminative AI models differ from generative AI models. Condition assessment uses discriminative models which assign input data to specific predetermined categories. Large language models together with diffusion-based image synthesis systems function as generative models which create synthetic data to boost training resources thus helping to solve the issue of insufficient data for uncommon defect types. The industrial inspection field currently uses generative models to develop new applications, but this area has potential for future research while working toward operational results.

4.2. Sociotechnical Systems (STS) Theory

The framework of Sociotechnical Systems (STS) theory explains how technical systems and human work practices function together to create operational systems within organizations (Trist & Bamforth, 1951; Trist, 1981). The theory states that technology requires study together with its complete sociotechnical system, which comprises users, operational methods, work habits, and organizational frameworks. STS theory establishes joint optimization as its fundamental principle because this principle requires technical and social system development to proceed simultaneously while both systems maintain shared understanding, yet the social system needs to adjust to the technical solution. This principle explains how AI should be adopted for condition assessment because AI tools need complete data-collection systems about role definitions and coordination structures to reach their objectives according to their technical standards. (Raisch & Krakowski, Artificial intelligence and management: The automation–augmentation paradox, 2021) established the automation–augmentation spectrum which Figure 3 displays while evaluating condition assessment within that spectrum. The figure shows that condition assessment exists in the augmentation zone because it needs AI to assist with decision-making tasks which require high tacit knowledge and unpredictable physical inputs and critical routing choices. The current setup determines how organizations need to develop AI tools and implement them inside their current operational procedures.

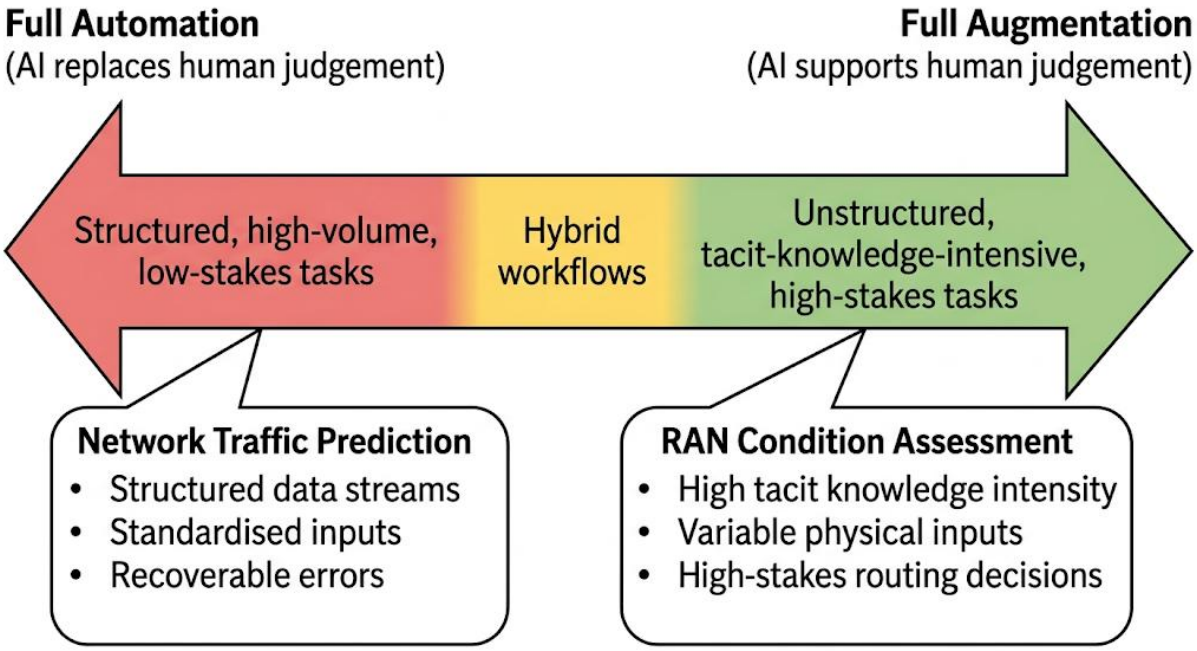


Figure 3 Automation–augmentation spectrum (adapted from Raisch & Krakowski, 2021).

The evaluation of reverse logistics operations establishes a direct connection to STS theory because the assessment process depends on human behavior to execute its tasks. The inspection process depends on two essential components, which are the unspoken knowledge and the professional skills acquired by technicians throughout their extensive years of experience, and which they find challenging to explain or document. Teams develop their operational methods through local customizations and resource limitations and informal practices, which exist separate from centralized control of operations. The decision-making process occurs through practical elements, which include time constraints and available resources and the uncertain detection of operational failures, but these elements fail to create clear structured data. The implementation of AI technology in this setting demands complete transformation of work functions and operational frameworks, which will reshape both work execution methods and standards of expert evaluation and decision-making processes.

Resistance to AI adoption can be understood through the lens of STS theory. When practitioners perceive AI as a threat to their expertise, autonomy, or professional identity, they respond with resistance through three non-compliant methods which include technicians making decisions without AI assistance and technicians who record data through methods which do not comply with system requirements. This resistance proves challenging because its effects remain hidden in overall performance metrics while it gradually develops into a systematic divide between planned operations and actual work procedures. Organizations need to implement participatory design methods which empower essential workers to establish the operational requirements of AI tools through their input on system establishment and decision-making processes and human supervision methods. AI adoption presents an organizational and cultural challenge because organizations need to assess how AI systems will work with their current operational procedures and how employees will evaluate and trust AI results and how work processes should be developed through joint efforts instead of being implemented through complete automation.

4.3. Data Quality Framework

Data Quality Framework focuses on the characteristics that determine whether data is suitable for analysis and decision-making. Key dimensions include accuracy, completeness, consistency, timeliness, and accessibility (Wang & Strong, 1996). The term accuracy describes the degree to which data demonstrates the actual equipment state, but this requirement gets broken throughout reverse logistics because technicians enter defect descriptions that do not match the actual equipment problems. The two attributes of completeness and consistency decide if all records have been documented completely while their format remains consistent throughout various teams and systems which must be established before their application in training reliable models. The ability to retrieve data and execute necessary actions depends on two factors which create challenges when information exists across multiple outdated systems. AI model performance drops when any single dimension decreases performance but multiple dimensions decrease performance together because compounding effects make automated condition assessment impossible.

The five dimensions of data quality established by Wang and Strong in 1996 show their respective impacts on AI model feasibility within reverse-logistics operations (cf. Figure 4). The figure demonstrates how each AI pipeline dimension, which experiences degradation, results in a particular failure mode that occurs when training labels become biased and feature vectors become incomplete and training data becomes unavailable (Budach et al. 2022).

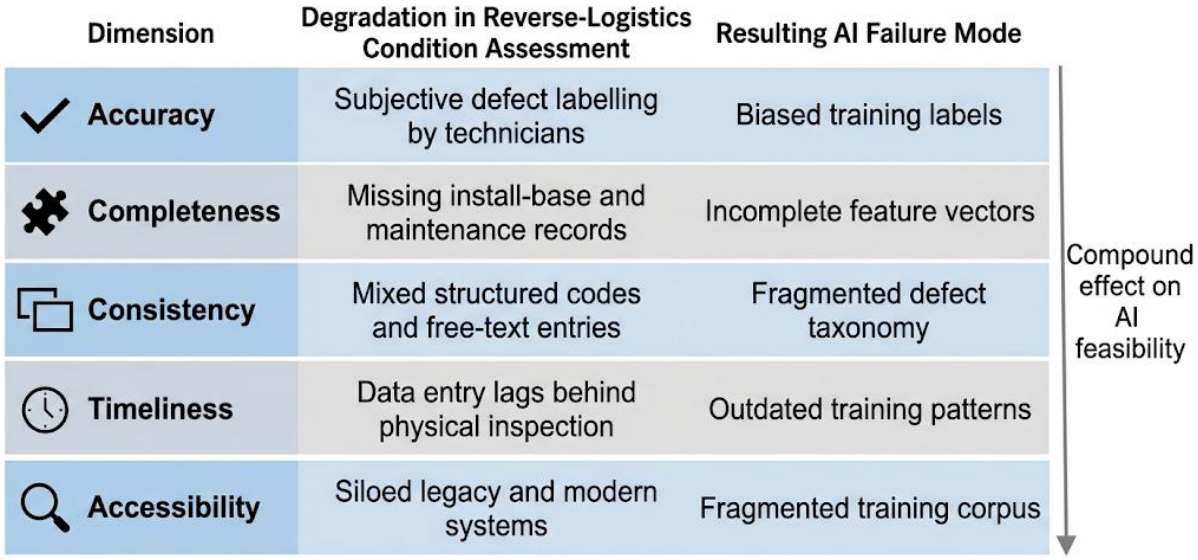


Figure 4 Data quality dimensions and their impact on AI feasibility (Wang & Strong, 1996).

The interconnections between all dimensions create negative effects which multiply their individual impacts on AI system development. Inaccurate labels reduce the signal available for model training; incomplete records reduce the coverage of the training dataset; inconsistent terminology fragments what would otherwise be a unified defect taxonomy into a collection of overlapping and ambiguous categories; untimely data decreases historical pattern relevance to present operational conditions; and poor accessibility prevents data from multiple systems from creating a unified training corpus. The empirical findings of this study demonstrate that when multiple dimensions experience simultaneous degradation in the reverse-logistics environment, the resulting conditions create an unsuitable data environment which requires governance intervention before organizations can use AI technologies.

The assessment process for reverse logistics requires accurate information. Returned equipment often arrives with incomplete or inconsistent documentation. Defect descriptions may rely on free text entries, technician judgement, or legacy systems that lack standardized taxonomies. The limitations of the system make it impossible to use AI based inspection tools because they need structured data to produce dependable results. Budach et al. (2022) provide empirical evidence that the relationship between data quality and model performance is non-linear: performance degrades slowly at first as quality declines from high levels but then drops sharply once quality falls below certain thresholds. The discovery found holds major significance because it helps determine which investments should receive priority during the current investigation. The data quality assessment shows that organizations need to reach a specific quality threshold before their performance enhancements from data quality upgrades will achieve substantial results. The non-linearity of AI value requires decision-makers to achieve complete data governance before they can observe AI benefits according to the business case assessment of reverse logistics AI implementation.

Data Quality Framework provides essential guidance that enables assessment of existing data environments which determine their suitability for AI adoption and their requirements for improvement. The system establishes an investment order by showing that data governance improvements need to happen before organizations can develop AI models which will proceed to train their models with high-quality data that will prevent the introduction of biases and errors from their data. Chapter 6 analytical framework and Chapter 7 recommendations both use this sequencing logic to present their findings.

4.4. AI Readiness and Maturity Frameworks

AI readiness frameworks assess an organization's capability to adopt and integrate AI technologies. These frameworks typically consider several dimensions (Alsheibani, 2018) (PricewaterhouseCoopers, 2018). Readiness assessment by these frameworks needs to be understood as multiple assessment dimensions instead of evaluating a single capability. The adoption of AI technology requires proper data infrastructure and governance system which enterprises need to establish before they can train their AI models and implement their digital workflows and process standardization needs to provide algorithms with consistent and structured data. The system requires more than just technical infrastructure because it needs employee competencies and technical skills to operate properly. The system needs technical skills and employee competencies because AI systems require operating staff to understand their results and manage their data pipelines. Organizational readiness to change their culture through innovation activities determines how employees and managers perceive AI while management dedication to AI projects drives their funding and organization-wide implementation. The entire process requires teams from different departments to work together because AI technology needs to operate in multiple operational departments which include reverse logistics and other logistic functions and repair and sustainability and data operations.

The dimensions of this system are interconnected because they create a network of abilities, whose weaknesses in one area block all progress toward other areas. The value of strong data infrastructure becomes limited when workflows do not meet the required level of standardization for producing uniform results. Organizations need their data environment to deliver essential backing for their employees' advanced skills to handle specific analytical tasks. The presence of management support needs cross-departmental cooperation because it produces standalone AI projects that lack the capacity to expand. Organizations must assess AI readiness through complete evaluation because existing system interdependencies create bottlenecks, which limit overall system development.

All existing reverse logistics operations depend on three factors which determine their AI readiness: the development level of their data collection systems, their operational procedures, and their capacity to embed sustainability goals into their business activities. (Jöhnk, 2021) discovered that organizations in operationally complex environments which have physical variations and need tacit knowledge and deal with multiple data ownership issues, tend to underestimate their current readiness levels. The telecom aftermarket context shows this finding, because network operations provide data-rich environments while reverse logistics operations create data-poor environments, which leads organizations to believe that their actual readiness for technology transfer exceeds their real capabilities. AI readiness frameworks help identify gaps between current practices and the conditions required for successful AI adoption. The empirical research results show organizational factors which the framework uses to create its structured approach. Chapter 6 analysis used AI readiness dimensions to assess the current organizational condition while determining the main required actions which included data governance and workflow standardization and competency development and governance alignment for reaching sustainable AI deployment readiness needed in condition assessment.

The study does not assess individual AI models, yet the research requires a basic classification system for its methods to explain how RAN condition assessment operations define "AI readiness." Different AI methods require different basic requirements which include specific data structures and labeling standards and requirements for maintaining time-based data integrity. The assessment of which techniques can be used with data conditions requires organizational and economic analysis instead of technical assessment. Table 2 shows which AI techniques are most suitable for reverse logistics condition assessment inspection tasks while it connects each technique to the data quality dimensions which Wang and Strong identified in their 1996 research. The mapping shows that

different techniques establish different readiness thresholds because the order of AI adoption funding decisions holds equal importance with technology selection.

Table 2 AI techniques applicable to RAN reverse-logistics condition assessment.

AI Technique	Applicable Task in RAN Assessment	Key Data Prerequisite	Wang & Strong (1996) Dimension at Risk	Representative References
Convolutional Neural Networks (CNN)	Visual defect detection: physical damage, corrosion, connector wear	Large, labelled image dataset	Completeness, Accuracy	(Cumbajin E, 2023); Gómez-Sirvent et al. (2022)
Gradient Boosting / Random Forest	Fault classification from structured repair logs and fault codes	Clean, structured tabular data	Consistency, Accuracy	(Mingyu, et al., 2024) (Marzuki, 2023)
NLP - Transformer-based (e.g., BERT)	Extracting condition signals from free-text technician notes	Unstructured text; no prior labelling strictly required	Interpretability, Consistency	(Buettner, 2025); (Zhong, Jackson, West, & & Cosma, 2024)
Anomaly Detection (Isolation Forest, Autoencoder)	Flagging out-of-distribution units without labelled failure data	Baseline normal-condition data	Completeness, Timeliness	(Zhang, Zhao, Dhupia, & & Xu, 2026)
LSTM Networks	Remaining useful life (RUL) estimation from temporal sensor sequences	Continuous, time-stamped operational metrics	Timeliness, Consistency	(Zhi-hong, Qing, Le-hao, & & Jing-jiao, 2021)

The table shows a specific trend which demonstrates that techniques which process unstructured data and low-label data work better with unstructured data because they need less data readiness

before they can be used in the organization which is being studied. The empirical findings from this study demonstrate that the required labelled datasets and continuous temporal data for CNN-based visual inspection and LSTM-based RUL estimation are currently not available [refer to Ch. 6 findings]. The existing distinction between two elements leads to the development of sequencing logic in the Discussion (Chapter 7) while proving the central argument which states that AI adoption decisions function as staged resource-allocation choices instead of operating as binary technology switches according to the Industrial Economics framework which has been adopted in this research.

4.5. Circular Economy and Sustainability Principles

The principles of circular economy focus on efficient resource management which includes material reuse and equipment repair activities and methods for decreasing waste production. The telecom industry requires vendors to extend their product life spans while decreasing their environmental effects and meeting sustainability certification standards (Ellen MacArthur Foundation, 2013) (Geissdoerfer, 2017). Condition assessment plays a central role in determining whether equipment can be reused or must be recycled, making it a key component of circular economy strategies. The reverse-logistics operation achieves its environmental and economic results through two factors which determine how routing decisions are made and how returned units are sorted into different disposal methods which include reuse and refurbishment and repair and recycling. Recycling operations that receive units from collection systems create material waste because they process items which should have been sent to other destinations and the process releases carbon emissions into the environment while units sent to reuse create reliability issues that lead to failures. The two misclassification types produce expenses which circular-economy principles aim to reduce; therefore, assessment accuracy directly affects sustainability performance.

The circular-economy disposition hierarchy which connects to returned RAN equipment shows its design through Figure 5 which extends from the Ellen MacArthur Foundation's ReSOLVE framework. The hierarchy ranks routing decisions by retained value and environmental impact, with direct reuse preserving the highest proportion of embedded resources and disposal representing total value loss. The assessment of accurate condition determines the decision point which decides the pathway each returned unit will take; the two errors which exist in this process create economic and environmental costs that AI-supported inspection is designed to minimize.

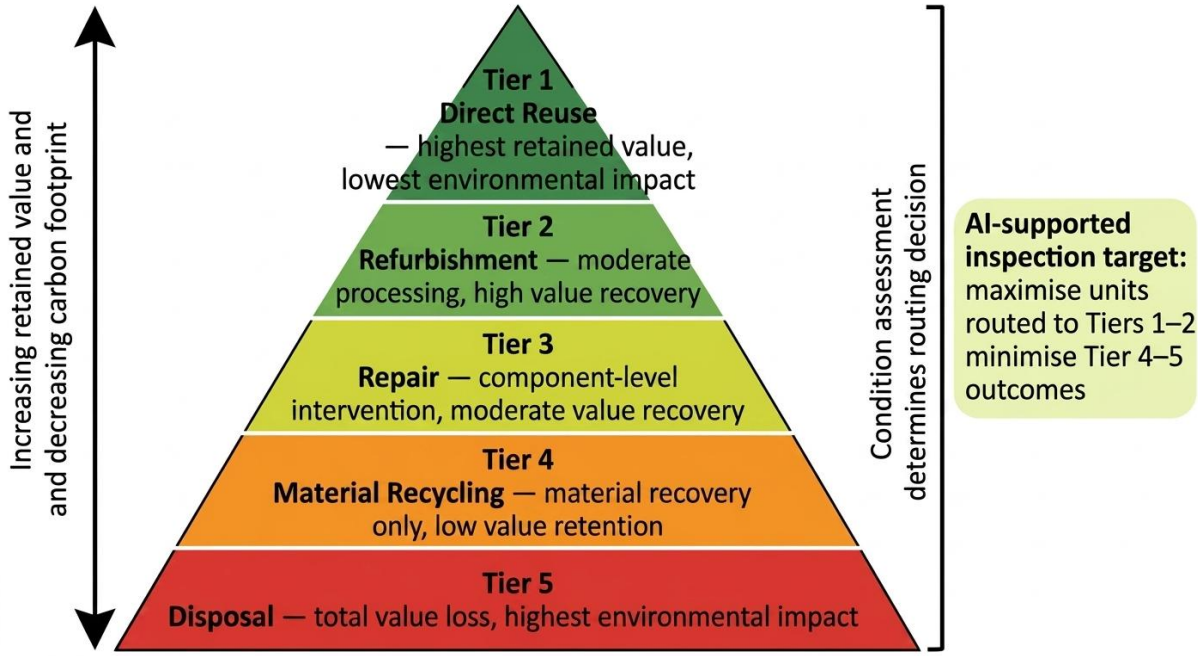


Figure 5 Circular economy disposition hierarchy for returned RAN equipment.

The research conducted by Saar and Thomas in 2003 established new circular economy principles which require products to control their own lifecycle through built-in information technology. The use of bar codes and RFID tags enables continuous tracking of product identity and composition and usage records which allow automatic product sorting and disassembly directions and recycling processes that provide rewards to users when products reach their end-of-life stage. The framework applied to returned RAN equipment shows a primary problem with data origin because manufacturers who do not create organized machine-readable data at the production site force reverse logistics operators to reconstruct that information through expensive methods which produce inconsistent results. The current study presents evidence of scattered metadata which matches the fragmented metadata environment shown in its empirical findings. The proposition "the products themselves will need to help us take care of them" from Saar and Thomas shows that AI systems face functional constraints which stem from SQ1 because AI models used for condition assessment require upstream data that should exist from the design stage.

The sustainability goals of a project determine whether an organization can successfully implement artificial intelligence technology. Circular-economy targets create direct incentives which improve condition assessment accuracy together with assessment consistency because high reuse rates depend on classification decisions which must remain both reliable and traceable. Organizations must fulfill the traceability requirement while establishing data governance processes which require them to record assessment methods and distribution paths for each unit because this requirement creates a need for standardized data systems which AI systems need to function properly. Organizations that invest in sustainability reporting and AI data readiness will experience combined benefits because these two areas need to work together instead of functioning as separate priorities. The European Union Corporate Sustainability Reporting Directive (CSRD) and Ecodesign Regulation requirements create additional urgent compliance obligations for organizations which need to present auditable proof of their reuse rates together with material recovery results and carbon footprint reductions. Organizations need to implement systematic assessment procedures which receive support from artificial intelligence technology because these procedures create accountability requirements which lead to better assessment processes.

As organizations investigate AI-supported inspection tools, circular economy principles provide an essential framework which helps to explain their research activities and their research limitations. The sustainability dimension creates additional complexity for AI adoption frameworks because AI systems must function as both operational decision-support systems and sustainability governance tools which produce traceable records to prove regulatory compliance. The dual requirements of operational efficiency and regulatory traceability create design challenges for AI systems which need to meet higher standards of success in this specific application area.

4.6. Summary

The theories presented in this chapter offer complementary perspectives on AI adoption in reverse logistics condition assessment. The technological foundation of Artificial Intelligence functions as the basis which Sociotechnical Systems theory describes how technology interacts with human work methods while Data Quality Framework demonstrates that accurate data serves as essential information. The frameworks provide guidance for Chapter 6 analysis which helps to create a comprehensive understanding of the research results. The four analytical frameworks are not merely additive; they are structurally interconnected in ways that reflect the multi-layered nature of the research problem. STS theory and Data Quality Framework share an explanatory claim: that data problems in condition assessment are not purely technical but are rooted in social practices, organizational routines, and tacit knowledge that resist straightforward standardization. AI Readiness frameworks and Circular Economy principles share a prescriptive orientation: both points toward the organizational investments, in governance, infrastructure, and strategic alignment, that would be required to advance AI feasibility while simultaneously meeting sustainability obligations. The combined application of all four frameworks therefore enables an analysis that is simultaneously diagnostic, explanatory, and prescriptive, addressing not only what the barriers to AI adoption are but why they exist and what would need to change to overcome them.

The relationship between the four theoretical frameworks used in this study and the three analytical dimensions of the research question is shown in Figure 6. The frameworks operate through three different connection methods because they create structural links between their different elements. The two frameworks establish a direct relationship between their social explanations of data issues because STS theory and Data Quality Framework both provide social origin explanations. The two frameworks establish a unified approach to examining governance and investment through their shared prescriptive practices. The empirical findings are interpreted through an integrated lens which consists of these two elements.

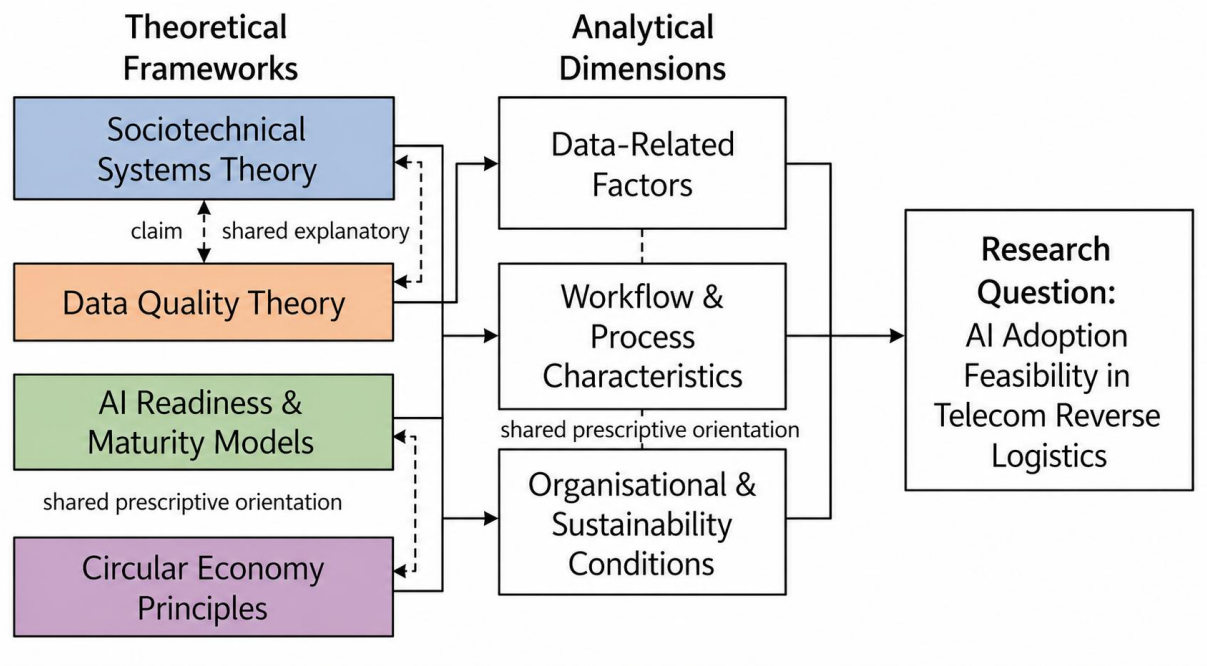


Figure 6 Integrated theoretical framework.

4.7. Application of Theoretical Frameworks in the Analysis

The Industrial Economics concepts introduced in Section 2.1.1, Transaction Cost Economics, the Resource-Based View, and Value Chain Analysis, provide the economic rationale for why AI adoption matters in reverse-logistics condition assessment. They frame AI implementation as a question of resource allocation, governance costs, and value creation. In contrast, the four analytical perspectives used in this chapter, Sociotechnical Systems (STS), the Data Quality Framework, AI Readiness and Maturity Models, and Circular Economy principles, guide the empirical interpretation of how AI adoption unfolds in practice. Together, these perspectives form a coherent multi-level analytical structure: Industrial Economics explains when AI adoption is economically viable, while the four analytical frameworks explain why adoption is challenging and how organizational, technical, and sustainability factors shape feasibility.

The four analytical perspectives outlined in this chapter provide complementary lenses for interpreting the empirical findings of this study. Together, they highlight different aspects of condition assessment and demonstrate how technical, organizational, and sustainability needs impact the success of AI implementation in telecom reverse logistics.

The research will use Sociotechnical Systems (STS) theory to study the impact of human judgment and tacit knowledge and existing work practices on AI integration with inspection tools. The interviews show that technicians use their practical experience to interpret physical damage and evaluate unclear fault descriptions and determine which tasks to perform. STS explains why human centered work methods result in different work patterns and why organizational processes need to match AI systems instead of being forced on workers. This lens helps to study how manual inspection work interacts with new digital inspection technologies. The study uses this framework to examine how practitioners respond to AI, including their trust levels, resistance, ability to adapt, and how current work processes may need to change to accommodate AI-generated outputs.

The Data Quality Framework will assist researchers in understanding the problems which arise from inconsistent defect descriptions together with missing metadata and incomplete install base information and excessive No Fault Found NFF rates. The limitations of these data elements make it impossible to develop effective AI models which produce dependable results. The analysis will use Data Quality Framework to investigate how data accuracy and completeness and consistency and accessibility affect the technical and economic feasibility of AI condition assessment systems. The framework shows how organizations that possess incomplete and unreliable data sets face major obstacles which prevent them from achieving automation success. The study will use this framework to determine the existing data environment and compare it with the minimum quality standard established in academic research for AI implementation. The study will determine which governance measures and standardized defect taxonomies and mandatory metadata fields and system integration, and labeling protocols will most effectively reduce the existing gap.

The study employs AI Readiness and Maturity Models to evaluate organizational factors that either enable or restrict AI implementation. The empirical findings highlight challenges such as immature IT governance and fragmented systems and restricted internal AI capabilities and customer restrictions on data use. The AI readiness frameworks enable organizations to assess their existing problems through an evaluation of their digital infrastructure and data governance methods and employee skills and strategic corporate alignment. The framework enables researchers to explain the reasons behind lower AI adoption rates in reverse logistics compared to other telecom sectors which include research and development and network operations. The organization will use this system to establish its position on the maturity continuum while determining which capabilities need investment according to their interdependent priority sequence for the organization to reach a level that enables continuous AI deployment in condition assessment.

The research will also use Circular Economy and sustainability principles to assess how environmental goals affect condition assessment methods and AI implementation. The interviews demonstrate that CO_2 reduction targets together with scrap reduction initiatives and component level reuse have become critical factors that influence operational decision making. The need to meet sustainability requirements creates new reporting obligations and restrictions which determine the methods organizations use to gather data and design their operational processes. The framework shows how sustainability pressures can simultaneously drive organizations to investigate AI solutions and complicate their implementation. The study will use the framework to examine how sustainability governance structures that include reporting obligations and reuse targets and compliance requirements affect organizations' decisions to adopt AI technologies and whether these structures serve as operational facilitators or limitations in the studied environment.

These four frameworks work together to create a unified approach for evaluating the collected empirical evidence. STS explains how people and their work activities interact with each other while Data Quality Framework establishes the limits of available information. AI Readiness models demonstrate organizational capabilities while Circular Economy principles provide a framework for understanding sustainability drivers. Their combined application enables a holistic interpretation of the factors influencing AI adoption in reverse logistics condition assessment. The frameworks are applied iteratively across the four empirical themes developed in Chapter 6, data-related factors, workflow and process characteristics, organizational and sociotechnical conditions, and sustainability-driven requirements, which enables the research question to be analyzed through its multiple research components using appropriate theoretical frameworks that maintain conceptual coherence throughout the entire chapter.

5. Method

This chapter describes the methodological approach used to investigate the feasibility of adopting AI-supported inspection methods in the condition assessment of returned Radio Access Network (RAN) equipment. The chapter outlines the research design, data-collection strategy, sampling approach, ethical considerations, analytical procedures, and measures taken to ensure credibility and validity. The aim is to provide transparency regarding how the empirical material was gathered, interpreted, and analyzed.

5.1. Data Collection Method

Given the exploratory nature of the research question and the limited prior empirical work on AI adoption in telecom reverse-logistics environments, a **qualitative research design** was selected. Qualitative methods are well suited for examining complex sociotechnical processes, uncovering tacit knowledge, and understanding how organizational routines, data practices, and sustainability requirements shape decision-making.

The primary data-collection method was **semi-structured interviews**. This format allowed for a balance between comparability across interviews and flexibility to explore context-specific insights. Interviews were conducted with participants from reverse logistics, supply-chain planning, refurbish and retake operations, field integration management, sustainability and circular-economy functions, network operations, procurement and warehousing, and AI-enabled network technology. These individuals were selected because they interact directly with returned-equipment workflows or contribute to data governance, digitalization, and decision-making processes relevant to condition assessment.

Due to privacy concerns and at the explicit request of the interviewees, **no audio or video recordings** were made during the interviews. Instead, **detailed notes were taken** during each interview session, and expanded transcripts were written immediately afterward based on these notes and the researcher's memory. This approach is common in qualitative research when participants prefer not to be recorded, and it aligns with ethical guidelines requiring respect for participant preferences. The eleven interviewees represented a diverse set of organizational contexts, including telecom equipment vendors, major telecom operators, and academic experts. Their roles spanned reverse logistics, supply-chain planning, refurbish and retake operations, field-integration management, sustainability and circular-economy governance, network operations, procurement and warehousing, and AI-enabled network technology across Europe, the USA, and APAC.

The interview questions aimed to investigate as much as possible factors influencing the feasibility of AI adoption in this environment. The study required participants to describe their current condition-assessment processes which included their actual decision-making methods and the specific points where their operations encountered problems. The research used questions about data accessibility and data quality and documentation methods because researchers needed to know which data existed and its accessible formats to determine whether AI could be implemented. The study required participants to describe sustainability-based reporting obligations which created both advantages and restrictions for their data management processes. The discussion moved to explore how organizations structure their operations to distribute information and expertise between different functions, while the study ended with an examination of how participants viewed their chances for using AI technology and their understanding of its restrictions based on their work experiences.

In addition to interviews, **internal documents** such as workflow descriptions, sustainability reports, and process guidelines were reviewed to support triangulation and strengthen the analysis.

5.1.1. Sampling

The process of selection of respondents was carried out using **purposive sampling techniques**. The intention was not one of representativeness but rather obtaining valuable insights from those who have experience of conducting a condition assessment and other organizational roles involved in the **reverse logistics chain**. Sample composition was done purposely to ensure that it **captured technical and organizational perspectives** as well. Participants in this study comprised individuals whose professions were reverse logistics, supply chain planning, refurbishing and retake activities, integration and decommissioning policy governance, sustainability and circular economy management, network operation, procurement and warehousing, and artificial intelligence-driven network technology.

The final **sample size** was guided by the principle of information saturation, where additional interviews no longer produced substantially new insights. To enable a 360-degree perspective, the sample included telecom vendors (Interviewees 1, 2, 3, 4, 8, 11), telecom operators (Interviewees 6, 7, 10), and academic experts (Interviewees 5, 9), allowing triangulation across organizational contexts.

5.1.2. Ethical Considerations

This research adhered to **ethical standards for qualitative research**. All participants were aware of the aim of the study and the fact that their involvement was voluntary and that they could withdraw from participation at any point. Written informed consent was sought before interviewing each participant.

Some participants **did not wish to be audio-recorded** because of confidentiality issues. This was agreed to, and extensive notes were made during interviews. All notes and expanded transcriptions were kept secure and were anonymized. There is nothing in the thesis which identifies the respondents, their positions in the organization, or the operational details of the firm. All the participants were assured that all their comments would remain anonymous and that none of the comments would be attributed to individuals. The research does not deal with sensitive personal information.

5.1.3. Overview of Interviews Conducted

A total of eleven interviews were conducted. The sampling strategy aimed to obtain a comprehensive, 360-degree view of the condition-assessment process by engaging participants from multiple organizational contexts and functional areas. Table 3 tabulates the interviewees information.

Table 3 Conducted interviews

S. No.	Role	Region	Company / Business Area	Duration & Mode
1	Capability Manager – Returns [Vendor]	Stockholm, Sweden	Global Telecom Equipment Manufacturer - Reverse Logistics & Returns Management	~60 min (MS Teams)
2	Material Planning Manager [Vendor]	Plano, USA	Global Telecom Equipment Manufacturer - Supply Chain, Repair Planning & Return Flow Coordination	~45 min (MS Teams)
3	Refurbish & Retake Supply Chain Manager [Vendor]	Europe	Global Telecom Equipment Manufacturer - Refurbish, Retake, Screening & Circular Economy Operations	~60 min (MS Teams)
4	Head of Field Integration Management – APAC [Vendor]	Asia Pacific Region	Global Telecom Equipment Manufacturer - Network Rollout, Dismantling, ASP/LSP Governance	~30 min (MS Teams)
5	Associate Professor – Business Management, Reverse Logistics [Academic]	Valencia, Spain	Polytechnic University of Valencia (UPV) - Reverse Logistics, Circular Economy, Decision Systems	~30 min (MS Teams)
6	Project Manager – Network Material & Procurement [Operator]	USA	Major Telecom Operator - Procurement, Reverse Logistics, Warehouse & Vendor Coordination	~45 min (MS Teams)

7	Senior Vice President – Network Technology & Operations [Operator]	Bonn, Germany	Major Telecom Operator - Network Operations, Reverse Logistics Governance, Sustainability	~45 min (MS Teams)
8	Group Lead – Environment & Climate [Operator]	Germany	Major Telecom Operator - Environment, Climate, Circular Economy, CSRD Reporting	~25 min (MS Teams)
9	Associate Professor – Circular Economy & Reverse Logistics [Academic]	Madrid, Spain	Technical University - Industrial Engineering, Circular Economy, Reverse Logistics	~30 min (MS Teams)
10	Network Inventory Manager [Operator]	Europe	Major Telecom Operator - Inventory, Warehousing, Returns, Reverse Logistics Coordination	~45 min (MS Teams)
11	Senior Executive – Telecom Networks, AI & Circular Economy [Vendor]	Global / Europe	Telecom Industry Leader - RAN, Open RAN, AI-enabled Operations, Circular Economy Strategy	~30 min (Google Meet)

5.2. Method of Analysis

The empirical data were analyzed using thematic analysis, a widely used method for identifying patterns within qualitative material. The process followed an iterative five-stage approach. First, the researchers familiarized themselves with the data through repeated readings of interview transcripts and supporting documents. Second, initial codes were generated by labelling segments related to data quality, workflow characteristics, sustainability requirements, and organizational conditions. Third, these codes were grouped into broader patterns that reflected themes occurring across multiple participants. Fourth, the themes were interpreted through the theoretical lenses outlined in Chapter 4, including Sociotechnical Systems theory, Data Quality Framework, AI readiness models, and Circular Economy principles. Finally, the themes were synthesized into the analytical framework presented in Chapter 6.

This approach enabled a systematic yet flexible analysis that remained grounded in the empirical material while informed by relevant theory. A detailed overview of the thematic structure, coding tables, and full narrative codebook is provided in Appendix B.

5.3. Credibility and Validity

Several methods were employed to boost the validity and credibility of the results obtained. Triangulation was achieved by comparing the information provided by the participants with that found in the internal documents, process maps, and sustainability reports. Reflexivity was exercised by reflecting on the influence that prior experience and assumptions of the researcher may have had on the results obtained.

The consistency of the analysis was secured by applying a well-designed coding scheme to all eleven interviews conducted. Detailed description of themes together with examples and quotations can be found in Chapter 6 of this paper.

Due to non-recording of the interviews, other methods of ensuring the quality of data were applied. Two researchers coded notes taken during the first interview and compared their findings. Themes emerged through analysis of several data sources rather than any individual statements made by the participants, which is typical of qualitative research.

5.3.1. Reflexivity and researcher bias

Qualitative research is inherently interpretive, and the researchers' prior experiences shape how interview material is collected, coded, and interpreted. To reduce the risk of biased interpretation, we explicitly discussed our own assumptions about AI, data quality, and circular economy prior to analysis and revisited these assumptions during coding meetings. We also treated themes as valid only when they appeared across multiple interviews and were supported by both interview notes and internal documents, rather than relying on single, striking quotes. This reflexive stance aimed to make the interpretive nature of the study transparent while limiting undue influence from our preconceptions.

The study design was informed by challenges identified in previous literature on AI adoption, data quality, and reverse logistics, such as fragmented data environments and workflow variability. Entering the field with these pre-identified issues created a risk of confirmation bias in both questioning and coding. To mitigate this, interview guides included open questions that allowed participants to introduce topics not anticipated from the literature, and the coding frame was developed inductively before being mapped onto theoretical constructs. Only after initial inductive coding did we align themes with the four analytical frameworks, which helped avoid forcing the empirical material into predefined categories.

5.4. Approach

The overall methodological approach can be summarized as exploratory, qualitative, and interpretive. It aims to examine how organizational processes, data environments, and sustainability considerations influence the feasibility of adopting AI in reverse-logistics operations.

Given the complexity and multidisciplinary nature of the problem, an exploratory qualitative design is appropriate. The research methodology was built on iterative analysis and theoretical interpretation.

The exploratory dimension focused on understanding the practical conditions for implementing AI technologies in reverse-logistics workflows. The qualitative dimension treated people and technology as an integrated sociotechnical system and involved examining work practices, processes, data flows, and organizational structures. The interpretive dimension consisted of several cycles of analysis aimed at identifying recurring patterns. The empirical material was interpreted using the theoretical perspectives described in Chapter 4, while remaining open to emergent themes not fully captured by existing theory, such as tacit knowledge transfer and operational barriers.

6. Analysis

This chapter presents the empirical findings derived from the interview material and supporting documents. The analysis is structured around four themes that emerged consistently across the dataset: **data-related factors**, **workflow and process characteristics**, **organizational and sociotechnical conditions**, and **sustainability-driven requirements**. These themes are interpreted through the analytical perspectives introduced in Chapter 4, including Sociotechnical Systems (STS) theory, the Data Quality Framework, AI readiness and maturity models, and Circular Economy principles. The purpose of this chapter is to explain how these themes shape the feasibility of AI-supported condition assessment in telecom reverse logistics. The detailed coding structure, including the theme hierarchy, coding tables, and narrative codebook, is provided in Appendix B.

6.1. Data-Related Factors

Data-related issues were the most frequently mentioned barriers across the interviews. Participants consistently described a fragmented information environment in which relevant assessment data exists but is not captured, structured, or integrated in a way that supports reliable AI development. This theme directly addresses in **first Sub-question (SQ1)**, which examines how data characteristics influence AI feasibility.

6.1.1. Data Availability and Fragmentation

A recurring finding across the interviews is that telecom reverse-logistics processes already generate a substantial amount of potentially relevant information, including serial numbers, defect descriptions, condition tags, images, test results, warehouse inspection notes, reasons for removal, and final disposition outcomes. However, the existence of such information does not by itself mean that it is analytically usable. Instead, the material suggests that the main problem lies in how these data are distributed across different operational systems, business functions, and process stages.

In 9 out of 11 interviews, participants explicitly referred to fragmented systems, dispersed records, or the lack of an integrated data view as a major obstacle to reliable assessment and future AI support. A procurement manager at a major US operator explained that serial numbers may be manually typed or photographed, technician notes are often free text, and documentation practices differ across regions and subcontractors, making the resulting data difficult to consolidate into a usable dataset. Similarly, a senior vice president at a major German operator stated that "the data exists, but it is not harmonized across systems," noting that field data, warehouse data, and vendor-return data remain stored in separate platforms with limited integration.

The same issue appears in a more strategic form in the interview with the senior executive focused on telecom networks and circular economy. He argued that the decisive problem is not merely the absence of records, but the absence of a reliable end-to-end data trail linking removal conditions, software state, integration history, and operational context to downstream reuse or disposal decisions. In his account, once this trail is missing, condition assessment becomes "largely guesswork," particularly in modern RAN environments where the condition of a unit is shaped not only by visible hardware damage but also by software configuration and system interaction.

These findings indicate that data-related barriers are not limited to missing fields or isolated documentation failures. Rather, they reflect a structurally fragmented information environment in which relevant data exists, but not in a form that supports robust AI training, validation, or decision support. Interpreted through the Data Quality Framework (Zainal, Baharudin, & & Khalid, 2024) (Xie, Sun, & & Zhao, 2025) (Budach, et al., 2022), the problem therefore concerns not only

completeness, but also consistency, accessibility, and contextual relevance. From an Industrial Economics perspective, such fragmentation increases coordination costs and information asymmetries, as organizations must invest substantial effort in locating, reconciling, and validating information before it can support economically meaningful automation. Selected visual outputs from the coding process (word cloud, treemap, bar charts, and cumulative code plots) are presented in Appendix A to illustrate the distribution of codes across the interview corpus.

6.1.2. Inconsistent labelling and Tacit Knowledge

A second recurring issue concerns the inconsistency of labelling practices and the continued dependence on tacit judgement in the interpretation of defects and condition states. Across the interviews, participants repeatedly emphasized that similar faults may be described in different ways depending on the technician, subcontractor, region, or product family involved.

In 8 out of 11 interviews, respondents referred directly to inconsistent terminology, unstructured defect descriptions, free-text entries, or the absence of standardized fault categories as a recurring problem in the assessment process. The US operator interview highlighted "inconsistent terminology across technicians and regions," "no standardized defect categories," and varying subcontractor practices as major contributors to No-Fault-Found disputes, credit delays, and uncertainty about reuse potential. The German operator similarly observed that many entries remain free text and that photographs vary widely in quality and angle, making analytical comparison difficult.

At the same time, the interview material shows that these data problems are closely linked to human expertise rather than simply poor discipline. The capability manager for returns described a more structured process involving predefined dropdown fault codes and mandatory images, yet still reported substantial variation in data quality and a No Fault Found rate of around 9 percent. This suggests that even where formal data fields exist, practical interpretation remains shaped by context and human judgement. The academic expert in reverse logistics reinforced this point by arguing that human error in manual entry is not incidental but structural, particularly when classification schemes are incomplete and heterogeneous items must be fitted into categories that do not adequately reflect field reality.

These observations show that inconsistent labelling is not merely a formatting problem. It reflects a deeper challenge of translating situated and experience-based knowledge into stable categories that can support automation. From a sociotechnical perspective, the issue lies in the relationship between human work and digital representation: field experience and local judgement remain central, but they are only partially codified into structured data. For AI adoption, this weakens the quality of labels used for supervised learning. For Industrial Economics, it indicates that codification itself is a costly organizational task, requiring investments in shared definitions, governance, and process redesign rather than only better software tools (Chowdhury, Budhwar, Dey, Joel-Edgar, & Abadie, 2022) (Gelashvili-Luik, 2025) (Jarrahi, Askay, Eshraghi, & Smith, 2023) (Mooradian, 2024).

6.1.3. Limited Historical Data for Rare Defects

A third data-related issue concerns the lack of historical and contextual information about the lifecycle of returned equipment. Several interviewees indicated that the physical appearance of a unit alone is often insufficient to determine its actual condition, reuse potential, or likely failure mode.

The senior executive argued that effective assessment should ideally be based on a fuller operational profile, including variables such as deployment duration, traffic load, energy consumption, number of power cycles, thermal stress, and software integration history. In his view, such information turns assessment from a static visual exercise into an interpretation of the unit's "life story," and this is especially important in telecom environments where faults may be linked to interaction effects rather than visible damage. His argument also challenges a narrow understanding of reverse-logistics data as mainly warehouse or inspection data. Instead, it suggests that feasibility depends on whether operational and network-level data can be connected to downstream reverse-logistics decisions.

Other interviewees echoed on the same problem from a more operational angle. Older equipment may have faded labels, missing parts, or incomplete records, while units removed under emergency conditions are often poorly documented because uptime restoration takes priority over careful capture of contextual data. The result is that organizations may hold partial records, but not sufficiently rich or reliable histories to support more advanced AI-supported classification.

This finding is important because it shows that AI feasibility is constrained not only by data quality in a narrow sense, but by the shallowness of the available representations of equipment condition. In Data Quality Framework terms, the issue affects representational adequacy as much as completeness. In Industrial Economics terms, the absence of context raises uncertainty about the real value of returned units and increases the risk of misclassification, thereby weakening the economic case for AI unless lifecycle data capture is improved in parallel.

The detailed codes supporting this theme are presented in Table B2.1 **in** Appendix B.

6.2. Workflow and Process Characteristics

The second theme concerns the workflow conditions under which returned equipment is removed, routed, inspected, and classified. Across the interviews, the process was described as functional in principle but highly variable in practice, shaped by time pressure, local judgement, and differences in regional or subcontractor execution. This theme is closely related to second Sub-question **(SQ2),** which addresses how current inspection routines, workflow variability, and manual decision-making affect the potential for AI integration.

6.2.1. High Variability in Returned Equipment

Eight interviewees described high variability in product age, configuration, dismantling quality, and documentation. Reverse logistics handles heterogeneous items under inconsistent conditions, making standardization difficult. Variability becomes problematic when combined with weak process discipline. A well-documented older unit may be easier to assess than a poorly documented newer one. Thus, the challenge is not heterogeneity alone, but heterogeneity without structured capture.

Practitioners gave more concrete illustrations of this heterogeneity. The operator interviews described older equipment with worn labels and missing components, units removed during emergency outages with incomplete documentation, and different product generations requiring

distinct inspection steps that field tools do not always support effectively. The returns manager also highlighted variation arising from incorrect or outdated configuration data and customer misperception of faults, both of which contribute to unnecessary returns.

At the same time, the senior executive offered a more nuanced interpretation by arguing that variability itself is not always the decisive blocker. In his account, an older unit that has been properly decommissioned and documented may be easier to assess than a newer unit removed hastily without structured data capture. This observation is analytically important because it suggests that the problem is not simply object heterogeneity, but the interaction between heterogeneity and process discipline. Variability becomes especially problematic when workflow design fails to generate stable, comparable, and context-rich observations (Masiha, 2021) (Tamang, R., Dazeley, R., & Aryal, 2025).

6.2.2. Manual Routines and Tacit Decision-Making

Nine interviewees emphasized that key assessment decisions rely on tacit judgement distinguishing cosmetic from functional damage, interpreting ambiguous notes, deciding whether to test a unit, or determining reuse potential. This reliance on tacit knowledge creates inconsistency and limits the transferability of expertise into algorithmic rules. From an **STS perspective**, condition assessment is a sociotechnical system where human judgement remains central. AI is therefore more likely to augment than replace expert work.

6.2.3. Lack of Standardized Workflows

Eight interviewees described insufficient standardization across warehouses, regions, and subcontractors. Even when formal procedures exist, execution varies due to time pressure, tool differences, and local routines.

The senior executive emphasized that structured data capture must be embedded into the physical workflow such as mandatory scans, guided apps, automated readings rather than added as an administrative task. Poor standardization increases coordination costs and weakens the business case for AI.

The detailed codes supporting this theme are presented in Table B2.2 in Appendix B.

6.3. Organizational and Sociotechnical Conditions

The third theme concerns the organizational conditions that shape readiness for AI adoption, especially cross-functional coordination, competence and capability distribution, and trust in AI-supported tools. This theme addresses third Sub-question (SQ3) by examining how governance structures, organizational maturity, and sustainability-related accountability affect the practical feasibility of AI adoption in reverse logistics.

6.3.1. Cross-Functional Coordination Challenges

Eight interviewees described fragmented ownership of the return process. Field operations, logistics, warehouses, procurement, engineering, finance, and sustainability all contribute, but no single function owns the end-to-end workflow.

This fragmentation complicates AI adoption because improvements in data capture, workflow design, and governance require coordinated action. From an **Industrial Economics** perspective, this is a coordination problem: costs and benefits are distributed unevenly across functions.

6.3.2. Skills and Competence Gaps

A second organizational issue concerns the maturity of the process and the capability base required to implement AI effectively. The interviews suggest that organizations are not operating from a blank slate. There is awareness of AI opportunities, some pilot projects, and strategic interest in digitalization. However, this interest coexists with significant gaps in process maturity, data readiness, and cross-functional capability (Verma, 2023; Dafallah, 2023).

Seven interviewees described early-stage AI maturity, weak data governance, and insufficient digital competence among frontline staff. Reverse logistics processes are less mature than forward logistics, making them weaker candidates for automation. Readiness is therefore not a static checklist but an emergent capability requiring shared understanding, stable decision criteria, and process maturity.

6.3.3. Trust and Acceptance of AI Tools

A third organizational pattern concerns trust and acceptance. Even interviewees who saw substantial potential for AI rarely described it as a substitute for expert judgement. Instead, they framed AI as a support mechanism that could improve validation, prioritization, anomaly detection, or early classification, while leaving humans responsible for higher-stakes or ambiguous decisions.

Six interviewees emphasized the need for interpretability and human oversight. AI is viewed as a support tool rather than a replacement for expert judgement. Trust depends on transparency, data lineage, and alignment with professional responsibility. From an **STS perspective**, trust is a sociotechnical property, not a technical one.

These observations reinforce the sociotechnical character of AI adoption (Benk, Ziegler, & Lütge, 2022) (Dorton, Korman, Ralph, & Speaker, 2024) (Golgeci, 2025). Trust is not produced by technical performance alone. It depends on whether AI can be embedded into a work system in a way that preserves accountability, supports interpretability, and aligns with how professionals understand their responsibilities. The implication is that successful implementation in this domain is likely to depend on hybrid human-AI configurations rather than full automation, at least in the medium term.

6.4. Sustainability-Driven Requirements

The fourth theme concerns the role of sustainability and circular-economy requirements in shaping both the pressure for improved condition assessment and the conditions under which such improvement must occur. Across the interviews, sustainability did not emerge as a peripheral concern. Rather, it appeared as a factor that alters the strategic and economic significance of classification quality, reuse decisions, and traceability.

6.4.1. Circular-Economy Targets and Reporting

Interviewees described a growing need to track reuse rates, recovery rates, scrap levels, waste diversion, and CO2 savings associated with repair and reuse decisions. These metrics increasingly feed into ESG reporting, internal circularity targets, and broader sustainability commitments.

In 7 out of 11 interviews, participants explicitly linked assessment quality to sustainability metrics, reporting obligations, or circular-economy targets. The German operator emphasized that

misclassification directly affects sustainability reporting. The US procurement manager similarly noted that if a unit is incorrectly classified as scrap when it could have been reused, the organization's reported sustainability performance is weakened. The returns capability manager added a more concrete business-case perspective, reporting that direct shipments between local companies could reduce lead time by 37.5 percent and cut CO2 emissions by 17 percent.

These findings indicate that condition assessment now has significance beyond operational routing efficiency. It also shapes the credibility of sustainability reporting and the organization's ability to demonstrate circular value creation. From a Circular Economy perspective, traceability and correct classification are therefore central enabling mechanisms for reuse and recovery, rather than secondary reporting functions. From an Industrial Economics perspective, sustainability requirements increase the economic relevance of getting classification right, because they affect not only cost and reuse rates but also regulatory credibility and strategic reputation.

6.4.2. Pressure to Increase Reuse and Reduce Waste

A second sustainability-related pattern concerns the growing pressure to extend equipment life, reduce scrap, and increase the share of units directed to reuse, refurbishment, or vendor take-back programs. However, the interviews also show that such ambitions are difficult to realize consistently when operational urgency dominates the practical workflow.

In 8 out of 11 interviews, participants referred to some form of tension between circularity goals and the immediate operational priorities of speed, uptime, or project delivery. Both major operators observed that careful dismantling and documentation often become secondary when projects are delayed or field teams are measured primarily on speed and delivery. The senior executive interpreted this as a design problem, arguing that circular practices must become the "path of least resistance" if they are to prevail in operational reality. The academic expert reinforced the same point by emphasizing design for repair, design for inspection, and design for multiple lifecycles as prerequisites for meaningful circularity.

The significance of this finding is that sustainability pressure does not automatically produce circular outcomes. Instead, it exposes the extent to which existing workflows still reflect a predominantly linear operational logic in which return, inspection, and reuse are secondary to faster restoration and delivery objectives. In this sense, sustainability operates both as a driver of change and as a diagnostic lens revealing the limitations of current process design.

6.4.3. Alignment Between Sustainability and Operational Priorities

A final pattern concerns the degree to which sustainability is embedded in operational governance and incentives. Although sustainability teams are increasingly active in promoting reuse, reporting, and circular programs, they often do not control the frontline workflow that determines how equipment is documented, inspected, and routed.

Six interviewees described misalignment between sustainability goals and operational KPIs. Sustainability teams promote reuse, but they do not control the workflow that determines documentation quality and routing decisions. This reflects an incentive-alignment problem: circularity goals are not embedded in operational governance.

These findings suggest that sustainability is strategically visible but only partially operationalized. In practical terms, the logic of circularity remains unevenly translated into incentives, accountabilities, and everyday routines. From an Industrial Economics perspective, this can be interpreted as an incentive-alignment problem: unless the functions responsible for dismantling, documentation,

inspection, and reuse decisions are rewarded in ways that support circular outcomes, sustainability objectives will remain weakly connected to operational execution.

The detailed codes supporting this theme are presented in Table B2.4 in Appendix B.

6.5. Cross-Theme Synthesis

Taken together, the four themes indicate that the feasibility of AI-supported condition assessment in telecom reverse logistics is not determined mainly by the existence of AI techniques, but by the maturity of the surrounding sociotechnical and organizational system. Data fragmentation weakens the reliability of assessment records, workflow variability prevents the consistent generation of high-quality inputs, fragmented governance obstructs coordinated process improvement, and sustainability demands increase the need for traceable and defensible classification.

The interview material therefore suggests that AI adoption should not be conceptualized as a narrow technology-implementation project. It is better understood as a multi-level organizational transformation that requires integrated improvements in data capture, workflow standardization, governance structures, and incentive design. Under current conditions, AI appears feasible primarily in support roles such as image-based pre-classification or anomaly flagging where data is relatively structured and human experts remain in the loop. Broader automation of condition assessment would require prior investments in integrated data trails, standardized workflows, and clearer end-to-end ownership of the return process. In this sense, the four themes collectively suggest that AI adoption is less a standalone technology project and more a multi-level organizational transformation.

This synthesis provides the foundation for the discussion in Chapter 7, where the implications of these findings are examined in relation to the theoretical frameworks and the main research question.

7. Discussion

This chapter discusses the findings in relation to the theoretical frameworks presented in Chapter 4 and the broader Industrial Economics lens introduced in Chapter 2. The discussion argues that AI adoption in telecom reverse logistics is constrained less by the absence of technical possibilities than by the interaction between weak data foundations, variable workflows, fragmented organizational coordination, and sustainability-related accountability requirements. In this sense, the study contributes not only to the understanding of AI adoption, but also to the understanding of how digital transformation unfolds in operational environments characterized by uncertainty, heterogeneity, and competing performance logics. The discussion builds on the empirical themes presented in Chapter 6 and the coding evidence provided in Appendix B.

7.1. Data-Related Factors and AI Feasibility

The findings show that data-related barriers to AI in telecom reverse logistics are systemic rather than incidental. Fragmented systems, inconsistent defect descriptions, missing contextual variables, and weak interoperability jointly reduce the quality of the information base required for reliable AI-supported assessment. The importance of this pattern is reinforced by the breadth of the sample: 9 out of 11 interviewees referred directly to fragmentation or the absence of an integrated data view, while 8 out of 11 highlighted inconsistent labels, free-text entries, or weak fault categorization as recurring problems.

Interpreted through the Data Quality Framework, these findings suggest that AI feasibility is undermined not by one isolated data-quality dimension, but by the simultaneous degradation of several dimensions at once. Accuracy is weakened by manual entry errors, consistency by divergent taxonomies and free-text practices, completeness by missing metadata and limited lifecycle information, and accessibility by the dispersion of relevant data across disconnected systems. This supports the view that data quality in operational AI contexts must be understood as relational and process-dependent, rather than as a narrow issue of cleansing a fixed dataset.

The findings also extend prior assumptions about data sufficiency by showing that the key issue is not only the quantity of available data, but the adequacy of the representation of equipment condition. In telecom reverse logistics, the condition of a returned unit may depend on traffic load, software state, power behavior, thermal exposure, and context, yet much of the available data remains detached from these operational dimensions. This creates a gap between what the AI model would ideally need to know and what the organization records.

From an Industrial Economics perspective, poor data quality increases information asymmetries and transaction costs throughout the reverse-logistics process. When firms cannot confidently assess whether a unit should be reused, repaired, returned, or scrapped, they incur additional costs through re-testing, disputes, delayed credits, unnecessary transport, and foregone value recovery. AI may promise efficiency gains, but those gains remain contingent on prior investments in data governance and integration. This implies that data quality should be viewed as a strategic capability and as a form of organizational infrastructure for future automation rather than as a secondary technical concern.

7.2. Workflow Variability and Process-Related Constraints

A second major implication of the findings is that workflow variability is not a peripheral complication, but a central determinant of AI feasibility. While the formal process for handling returned equipment may appear relatively straightforward, the actual performance of the process varies substantially across sites, subcontractors, product types, and operational circumstances. This

matters because AI adoption depends not only on having data, but on generating data through sufficiently stable and repeatable routines.

The empirical material shows that 8 out of 11 interviewees explicitly referred to variability or heterogeneity, and 9 out of 11 emphasized the role of human judgement or tacit knowledge in key decisions. These patterns indicate that the workflow has not yet been reduced to a set of standardized and fully codified decision rules. Instead, much of the work still depends on the situated expertise of technicians, inspectors, and managers who interpret incomplete data under time pressure.

This finding is highly consistent with Sociotechnical Systems theory. Condition assessment is not merely a technical act of identifying a defect. It is a sociotechnical process in which human experience, organizational incentives, digital tools, documentation practices, and local contingencies interact to shape outcomes. The continued importance of tacit knowledge suggests that the current environment is more compatible with augmentation than with full automation. AI can support triage, consistency checks, anomaly detection, or image validation, but it cannot yet replace the interpretive work that occurs when the process itself remains only partially standardized.

From an Industrial Economics perspective, unstable workflows weaken the business case for AI because they raise the cost of complementary investments needed to make automation effective. Organizations must not only develop or acquire AI tools; they must also standardize dismantling practices, improve documentation discipline, align local routines, and reduce variability in how cases are handled. These complementary investments are part of the real cost of adoption. The findings therefore help explain why AI can be seen as technically promising while still remaining operationally difficult to scale.

7.3. Organizational Readiness and Sociotechnical Alignment

The findings further indicate that organizational readiness for AI should be understood primarily as a governance and alignment problem rather than as a purely technical capability issue. Across the interviews, respondents described interest in AI, awareness of potential use cases, and in some cases ongoing pilot initiatives. However, they also described unclear process ownership, siloed systems, uneven competence distribution, and limited trust in automated decision support.

This pattern is again broad across the sample. In 8 out of 11 interviews, participants referred to fragmented ownership or cross-functional silos, while 7 out of 11 described their organizations as still requiring foundational improvements in data governance, process maturity, or AI-related capability before larger-scale adoption would be realistic. These results support the interpretation that readiness is not a binary state, but a cumulative organizational capacity built through governance clarity, shared process understanding, technical competence, and trust-building.

Read through AI readiness and maturity frameworks, the case organizations appear to occupy an intermediate position. Digital tools exist, leadership interest is present, and some AI use cases are already being explored. Yet the empirical material suggests that core readiness dimensions remain incomplete, including integrated data platforms, standardized process inputs, cross-functional governance, and operational acceptance of AI outputs. This indicates that the bottleneck is not innovation ambition, but an institutional ability to coordinate implementation across organizational boundaries.

Sociotechnical interpretation is equally important. AI does not enter a neutral environment; it becomes embedded in an existing pattern of roles, authority, expertise, and accountability. If these elements are misaligned, technical feasibility alone will not produce adoption. In several interviews,

respondents implied that human experts must retain final authority where AI outputs remain difficult to interpret or justify. This suggests that trust in AI is shaped not only by model performance but by whether the organization has developed credible mechanisms for explanation, oversight, and integration into everyday decision-making.

From an Industrial Economics standpoint, fragmented governance creates underinvestment problems. The benefits of improved assessment are distributed across several functions, but the cost of change may fall unevenly across them. Under such conditions, no single unit may have sufficient incentive to bear the costs of redesigning the process, which helps explain why promising AI opportunities remain stuck in pilot stages. Organizational readiness is therefore also an incentive-design problem.

7.4. Sustainability-Driven Requirements and Circular-Economy Pressures

The study also shows that sustainability requirements play a dual role in adoption of AI. On the one hand, they increase the strategic importance of accurate condition assessment because reuse, refurbishment, recovery, and reporting all depend on credible classification. On the other hand, they expose the inadequacy of current data, workflow, and governance structures, since those structures often fail to produce the traceability and consistency needed for meaningful circular performance. The breadth of this theme is also visible in the interview material. In 7 out of 11 interviews, participants explicitly linked assessment quality to sustainability metrics or circularity targets, while 8 out of 11 described tensions between circular ambitions and operational urgency. These findings suggest that sustainability is not an external reporting layer added onto reverse logistics after the fact. It is increasingly part of the core value logic of the process itself.

Viewed through Circular Economy principles, the findings reveal a persistent gap between circular aspirations and operational execution. The academic expert emphasized the importance of design for repair, inspection, and multiple lifecycles, while the senior executive argued that circular practices must become the path of least resistance if they are to prevail under real operational conditions. Together, these perspectives suggest that circularity depends less on high-level ambition alone and more on whether organizations redesign processes, products, and incentives so that reuse becomes easier than disposal.

From an Industrial Economics perspective, sustainability changes the incentive structure of reverse logistics by increasing both the value of correct reuse decisions and the cost of poor classification. Yet the findings also show that sustainability remains only partially embedded in operational KPIs, especially for frontline teams. This means that the economic and environmental logic of circularity is not yet fully translated into the behavioral logic of daily work. AI adoption, in this context, may support circular-economy goals, but only if the organization aligns operational incentives with sustainability objectives rather than treating them as parallel agendas.

7.5. Addressing the Main Research Question

The main research question asked how data quality, workflow characteristics, and sustainability requirements influence the adoption of AI-based inspection techniques in the condition assessment of returned telecommunication equipment. The findings show that these factors do not operate independently. Instead, they interact through organizational readiness and governance to shape whether AI adoption becomes feasible, credible, and economically justifiable in practice.

First, the feasibility of AI is weakened by fragmented, inconsistently structured, and context-poor data. Second, workflow variability and reliance on tacit judgement prevent the generation of stable and repeatable inputs needed for reliable automation. Third, fragmented ownership and weak cross-functional governance reduce the ability of organizations to coordinate the complementary changes required across field operations, logistics, procurement, engineering, and sustainability. Fourth,

sustainability requirements intensify the need for more accurate and auditable classification, while simultaneously revealing how far current practices remain from supporting meaningful circularity.

The central implication is therefore that AI-supported condition assessment becomes viable not when a promising algorithm is identified, but when the surrounding organizational system reaches a sufficient level of maturity. That maturity includes integrated data governance, more standardized workflows, stronger end-to-end ownership, and incentive structures that align operational execution with circular value recovery. AI adoption in this domain is thus best understood as a staged sociotechnical and economic transformation rather than as a discrete IT investment decision.

7.6. Limitations and contextual scope

This study is based on a qualitative single-case design supported by semi-structured interviews with practitioners and experts across the telecom reverse-logistics ecosystem. It does not aim for statistical generalization. Instead, its purpose is to generate analytically transferable insights into the conditions that shape AI adoption in a complex operational setting.

Although the empirical core is grounded in a Nordic case, the interview material includes actors operating in broader European and international contexts, including large operators and strategic experts with multi-market experience. This broadens the relevance of the findings, particularly because the issues identified, fragmented legacy systems, weak interoperability, heterogeneous returned equipment, tacit assessment practices, and growing circular-economy pressures, are not unique to a single national setting. At the same time, the degree to which the findings transfer to other settings will depend on regulatory environments, digital maturity, contractual structures, product architectures, and the sophistication of local reverse-logistics operations.

A further limitation is that the study focuses on organizational and process preconditions rather than on benchmarking specific AI models or architectures. It therefore cannot determine which specific model would perform best for image inspection, anomaly detection, or classification. Its contribution lies instead in clarifying the organizational conditions under which such tools are more likely to succeed or fail.

7.7. Conclusion

This study shows that AI adoption in telecom reverse-logistics condition assessment is shaped by the interaction of data quality problems, workflow variability, organizational fragmentation, and sustainability-related pressures. The main barriers are not primarily algorithmic. Rather, they arise from the broader sociotechnical and economic conditions under which returned equipment is documented, interpreted, classified, and routed.

By combining Industrial Economics reasoning with Sociotechnical Systems theory, the Data Quality Framework, AI readiness perspectives, and Circular Economy principles, the study contributes a multi-level explanation of why AI adoption matters, why it remains difficult, and what must change for it to become viable. The findings suggest that organizations should prioritize data governance, workflow redesign, cross-functional ownership, and sustainability-aligned incentives before expecting substantial value from AI-supported condition assessment. In this respect, the path to AI in reverse logistics runs through organizational redesign as much as through technological innovation.

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Appendix A: QualCoder analysis outputs and coding overview

A.1 Coding approach and software

Interview transcripts were coded in QualCoder using an inductive, data-driven approach. Six main categories were developed: AI & Machine Learning, Condition Assessment Process, Data Quality & Management, Organizational Factors, Circular Economy & Sustainability, and Reverse Logistics Operations. These categories and their sub-codes were iteratively refined as new interviews were analyzed to ensure consistent application across the full dataset. Visual outputs from QualCoder were used to check the distribution of codes across cases and to verify that the coding scheme reflected the empirical material rather than preconceived expectations.

A.2 Visual Outputs

This appendix presents seven QualCoder visualizations used to support the cross-case analysis. Each figure is accompanied by a brief technical description and an explanation of its analytical function in the study. Together, these visuals provide a transparent account of how the coding structure was constructed and how thematic priorities emerged from the data.

A.2.1 Word cloud of the full interview corpus

Technical description: The word cloud (Figure 7) was generated from the plain-text content of all eleven interview files. Common stop-words were removed, and remaining words were rendered at a size proportional to their frequency.



Figure 7 QualCoder word cloud of the complete interview corpus.

Analytical function: The word cloud provides a language-level overview of the empirical material before any coding categories are applied. For example, the prominence of terms such as “unit”,

"parts", and "workflow" confirms that the discourse is grounded in physical equipment handling and process management rather than abstract technology discussion. The appearance of "human", "judgement", and "tacit" highlights the centrality of manual, expertise-dependent inspection routines in the dataset.

A.2.2 Treemap of categories and codes

Technical description: The treemap (Figure 8) is generated from the aggregate code-count report. Each coloured block represents a category and its component sub-codes, with area scaled to total occurrence count.

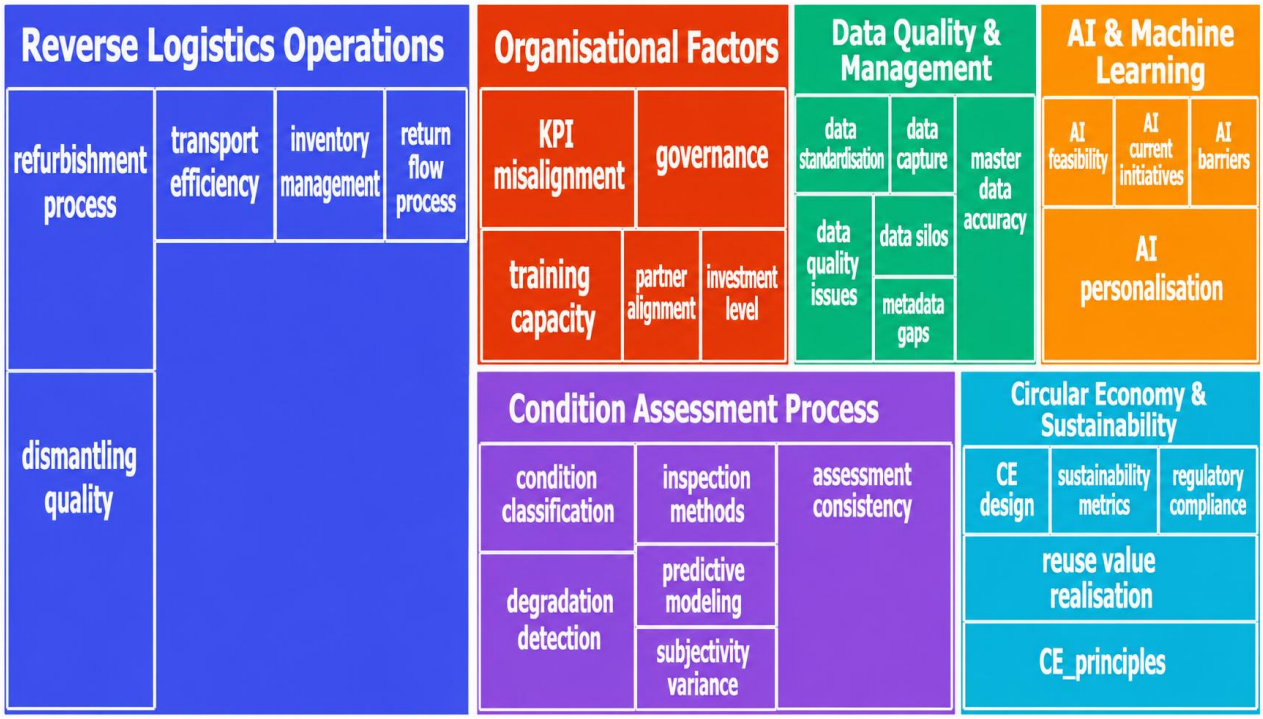


Figure 8 QualCoder treemap of code counts across all eleven interviews.

Analytical function: The treemap offers a compact view of how analytical weight is distributed across the coding scheme. It shows, for instance, that Reverse Logistics Operations and Organizational Factors occupy the largest areas, while AI & Machine Learning appears as a smaller but distinct category. This confirms that AI-related themes are embedded within a broader operational and governance context rather than dominating the empirical material.

A.2.3 Bar chart of code frequency

Technical description: The bar chart (Figure 9) displays the total count of each code applied across the full dataset in descending order.

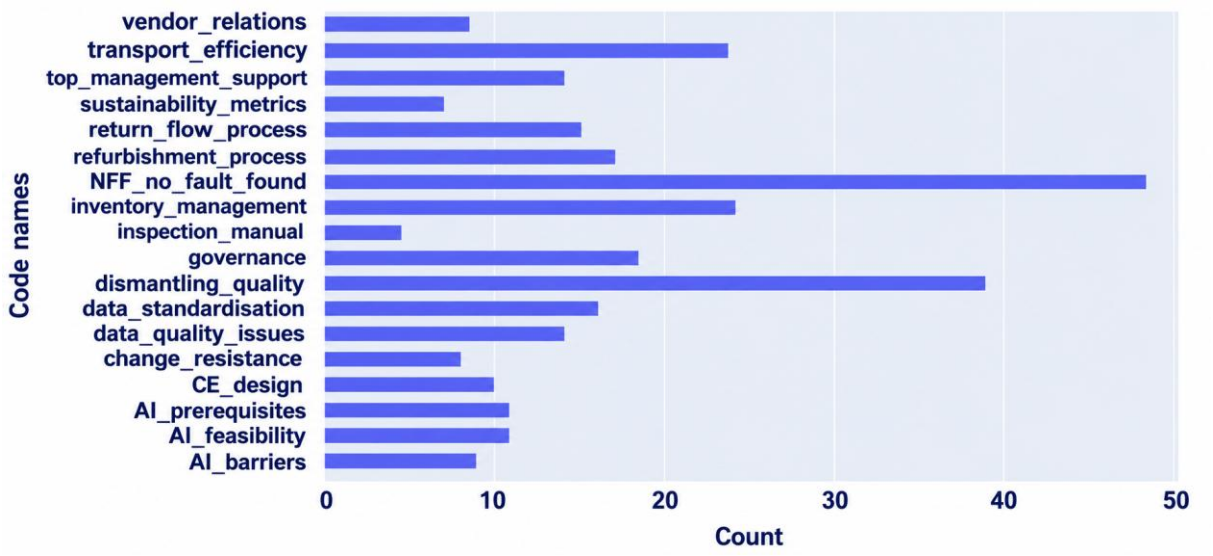


Figure 9 QualCoder bar chart of code counts across all eleven interviews.

Analytical function: The bar chart provides a granular quantitative view of coding density. It highlights which codes were most frequently applied and which remained more marginal, supporting transparent prioritisation of themes in the main analysis chapters.

A.2.4 Cumulative code count by case

Technical description: The cumulative case chart (Figure 10) aggregates the total number of coded segments per interviewee. Each bar represents one case, with coloured segments corresponding to individual codes.

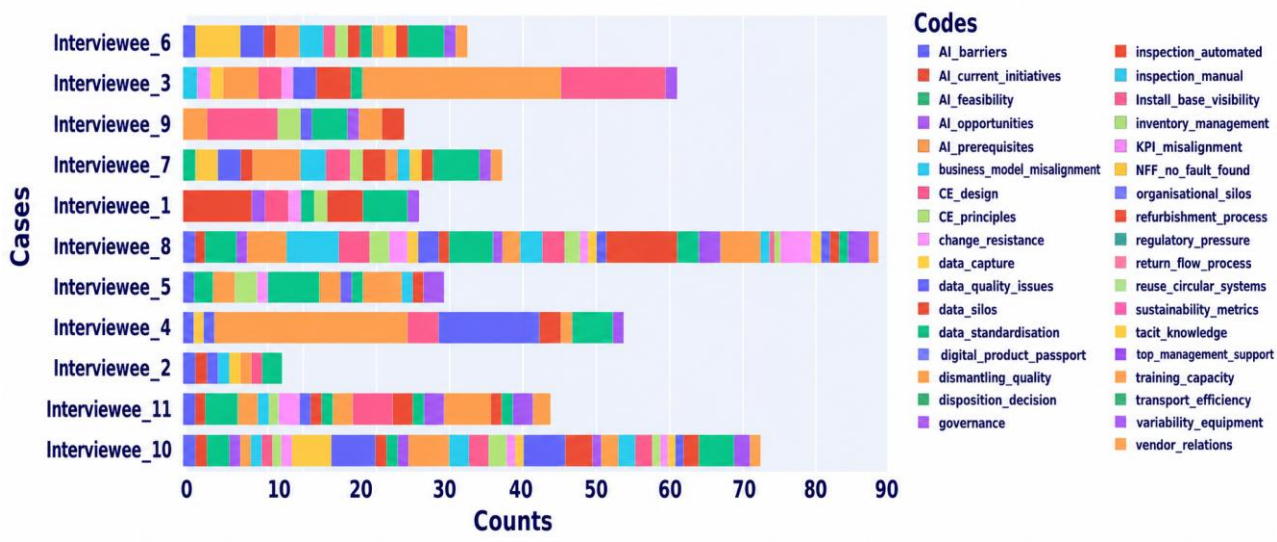


Figure 10 QualCoder cumulative code count by case.

Analytical function: This chart shows variation in coding density across interviewees and indicates which cases contributed the broadest thematic coverage. It is used to understand the relative breadth of each participant's perspective rather than to rank their importance.

A.2.5 Cumulative code count by file and code

Technical description: This chart (Figure 11) reverses the perspective of Figure A4 by aggregating files per code. Each bar represents a code, with coloured segments indicating which interviews contributed to that code.

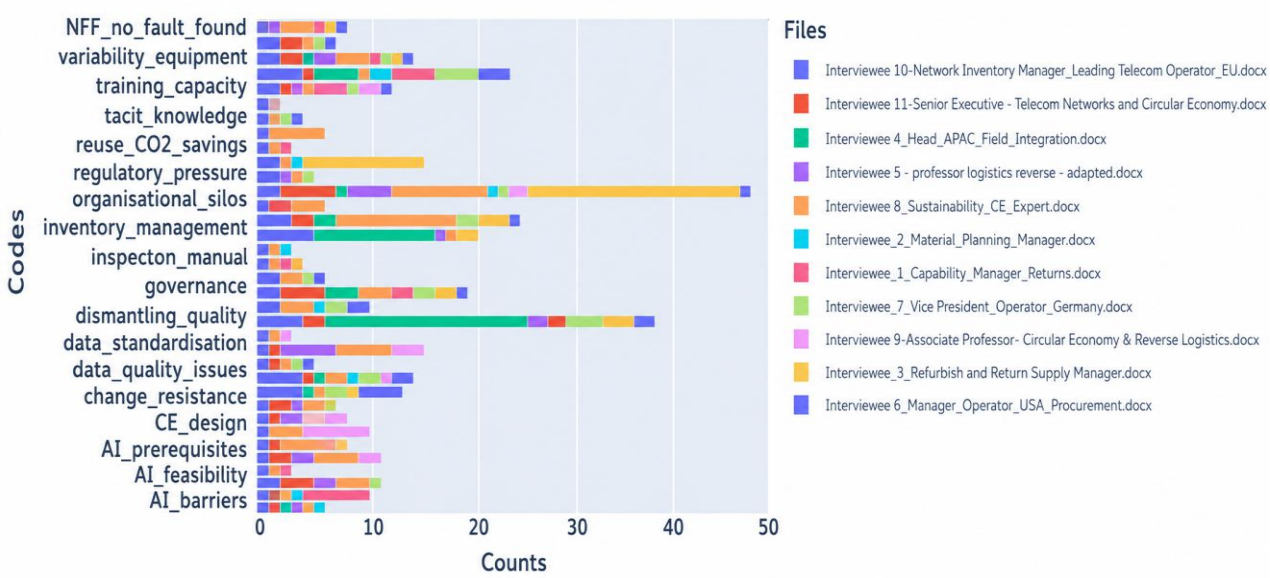


Figure 11 QualCoder cumulative code count by file and code.

Analytical function: The chart is used to distinguish between codes that are concentrated in one or two interviews and those that are distributed across the sample. This supports claims about whether a theme is systemic or case-specific.

A.2.6 Sunburst chart for the AI & Machine Learning category

Technical description: The sunburst chart (Figure 12) focuses on the AI & Machine Learning category. The inner ring represents the category, and the outer ring divides the total count across its constituent codes.

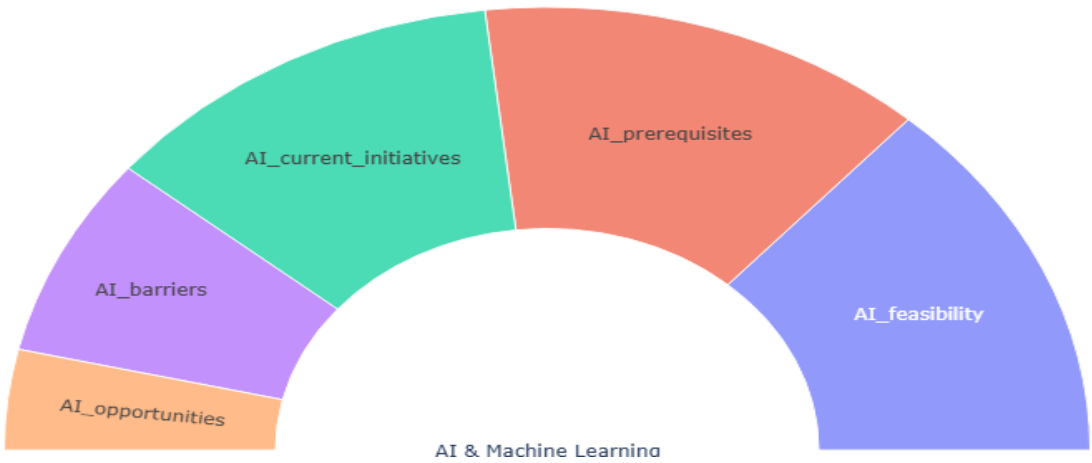


Figure 13 QualCoder sunburst chart for the AI & Machine Learning category.

Analytical function: The sunburst chart summarises how AI-related discussion is distributed across sub-themes such as feasibility, prerequisites, current initiatives, barriers, and opportunities. It is used to justify the relative emphasis placed on these sub-themes in the analysis chapter.

A.2.7 Case heatmap for the AI & Machine Learning category

Technical description: The heatmap (Figure 13) cross-tabulates interviewees (rows) against AI-related codes (columns). Cell colour intensity reflects the number of times a given code was applied to a given case.



Figure 14 QualCoder case heatmap for the AI & Machine Learning category.

Analytical function: The heatmap provides a compact overview of which AI-related themes were discussed by which interviewees and with what intensity. It supports cross-case comparison and helps identify where perspectives converge or diverge on AI feasibility and prerequisites.

Appendix B – Thematic analysis and codebook

This appendix presents the full analytical material supporting the thematic analysis described in Chapter 6. It includes the hierarchical structure of themes, the coding tables used to organize empirical patterns, and the complete narrative codebook. These materials provide transparency regarding how the empirical findings were derived from the interview data.

B.1 Theme Hierarchy Diagram

The thematic structure developed through inductive coding consists of four main themes, each containing several sub-themes (codes). The hierarchy is shown below:

Theme 1 - Data-Related Factors

1.1 Data Fragmentation
1.2 Missing Metadata
1.3 Inconsistent Defect Labels
1.4 Serial Number Errors
1.5 Poor Photo Documentation
1.6 NFF-Related Data Gaps
1.7 Install-Base Visibility Issues
1.8 Customer Restrictions on Data Sharing

Theme 2 - Workflow & Process Characteristics

2.1 High Variability in Dismantling Practices
2.2 Subjective Visual Inspection
2.3 Tacit Knowledge Dependence
2.4 Inconsistent Photo Capture
2.5 Non-Standardized Workflows
2.6 Time Pressure at Field Sites
2.7 Lack of Reference Images
2.8 ASP–LSP Disputes

Theme 3 - Organizational & Sociotechnical Conditions

3.1 Fragmented Ownership of Return Process
3.2 Siloed Systems and Data
3.3 Limited AI Maturity
3.4 Lack of Training & Digital Competence
3.5 Governance Gaps
3.6 Resistance to Change
3.7 Misaligned KPIs
3.8 Customer Restrictions on AI Use

Theme 4 - Sustainability-Driven Requirements

4.1 Reuse & Scrap-Reduction Targets
4.2 CO_2 Reporting Dependencies
4.3 Circular Economy Incentives
4.4 Conflicts Between Speed & Sustainability

4.5 Lack of Sustainability KPIs
4.6 Component-Level Reuse Opportunities
4.7 Regulatory Pressure (CSRD, EU)
4.8 Misalignment Between Sustainability & Operations

B.2 Coding Tables

The following tables summarize the codes used within each theme, including definitions, paraphrased example quotes, and links to the corresponding sub-questions and theoretical frameworks.

B2.1 Table - Theme 1: Data-Related Factors

(SQ1 + Data Quality Framework)

Theme	Code	Definition	Example Quote (Paraphrased)	Linked SQ	Linked Theory
Data-Related Factors	**Data fragmentation**	Assessment data is scattered across field apps, warehouse tools, vendor portals, and spreadsheets with no unified view.	"Information from field teams, warehouses, and vendors sits in different systems, so we never see the full picture."	**SQ1**	**Data Quality Framework**
Data-Related Factors	**Missing metadata**	Key fields (reason for removal, configuration, software version, accessories) are often absent.	"Units arrive with half the fields empty, no reason code, no context, nothing."	**SQ1**	**Data Quality Framework**
Data-Related Factors	**Inconsistent defect labels**	Technicians use free-text descriptions with no standard taxonomy.	"The same defect can be described in ten different ways depending on who typed it."	**SQ1**	**Data Quality Framework**
Data-Related Factors	**Serial number errors**	Manual entry leads to incorrect or unreadable serial numbers.	"We constantly receive units with wrong or unreadable serials because they were typed manually."	**SQ1**	**Data Quality Framework**
Data-Related Factors	**Poor photo documentation**	Photos vary in angle, lighting, and completeness,	"Photos are often blurry or don't	**SQ1**	**Data Quality Framework**

Theme	Code	Definition	Example Quote (Paraphrased)	Linked SQ	Linked Theory
		limiting AI training potential.	show the damaged area clearly."		
Data-Related Factors	**NFF-related data gaps**	Lack of structured failure data leads to high No-Fault-Found rates.	"Many units marked faulty pass testing because the original fault wasn't documented properly."	**SQ1**	**Data Quality Framework**
Data-Related Factors	**Install-base visibility issues**	Operators and vendors lack accurate, real-time install-base data.	"We don't always know what's actually installed at customer sites, the data is incomplete."	**SQ1**	**Data Quality Framework**
Data-Related Factors	**Customer restrictions on data sharing**	Some customers prohibit using their data for AI, limiting dataset completeness.	"Certain customers forbid us from using their data in AI systems, which blocks model training."	**SQ1**	**Data Quality Framework**

B2.2 Table - Theme 2: Workflow & Process Characteristics

(SQ2 + Sociotechnical Systems (STS) Perspective)

Theme	Code	Definition	Example Quote (Paraphrased)	Linked SQ	Linked Theory
Workflow & Process Characteristics	**High variability in dismantling practices**	Field teams dismantle equipment differently across regions, subcontractors, and customer programs.	"How equipment is removed depends entirely on the subcontractor and the site, no two teams do it the same way."	**SQ2**	**STS Perspective**
Workflow & Process Characteristics	**Subjective visual inspection**	Damage assessment relies heavily on technician	"Two inspectors can look at the same radio and reach different	SQ2	STS Perspective

Theme	Code	Definition	Example Quote (Paraphrased)	Linked SQ	Linked Theory
		judgement with no objective criteria.	conclusions about whether it's damaged."		
Workflow & Process Characteristics	**Tacit knowledge dependence**	Decisions rely on experience, intuition, and undocumented routines.	"A lot of decisions come down to what the technician 'feels' is wrong with the unit."	SQ2	STS Perspective
Workflow & Process Characteristics	**Inconsistent photo capture**	Photos vary in angle, lighting, and completeness, limiting standardization.	"Photos are taken differently at every site, some show everything, others show almost nothing."	SQ2	STS Perspective
Workflow & Process Characteristics	**Non-standardized workflows**	Inspection steps differ across warehouses, regions, and subcontractors.	"Each warehouse has its own way of doing inspections, there's no unified process."	SQ2	STS Perspective
Workflow & Process Characteristics	**Time pressure at field sites**	Technicians prioritize speed over documentation quality.	"When a site is behind schedule, documentation becomes the first thing to suffer."	SQ2	STS Perspective
Workflow & Process Characteristics	**Lack of reference images**	No golden samples or standard visual references exist for comparison.	"Inspectors rely on memory because we don't have reference images to compare against."	SQ2	STS Perspective
Workflow & Process Characteristics	**ASP–LSP disputes**	Disagreements occur about when and where damage happened.	"ASP says the unit was fine at handover, LSP says it arrived damaged, we	SQ2	STS Perspective

Theme	Code	Definition	Example Quote (Paraphrased)	Linked SQ	Linked Theory
			have no way to verify."		

B2.3 Table - Theme 3: Organizational & Sociotechnical Conditions

(SQ3 + AI Readiness & Maturity Models)

Theme	Code	Definition	Example Quote (Paraphrased)	Linked SQ	Linked Theory
Organizational & Sociotechnical Conditions	**Fragmented ownership of return process**	No single function owns the end-to-end return and assessment workflow.	"Field, logistics, warehouse, procurement, everyone owns a piece, but no one owns the whole process."	**SQ3**	**AI Readiness Models**
Organizational & Sociotechnical Conditions	**Siloed systems and data**	Field, warehouse, and vendor systems are disconnected.	"Our systems don't talk to each other, data sits in separate islands."	SQ3	AI Readiness Models
Organizational & Sociotechnical Conditions	**Limited AI maturity**	Organizations lack AI governance, skills, and infrastructure.	"We see the potential of AI, but we're not ready to support it operationally."	SQ3	AI Readiness Models
Organizational & Sociotechnical Conditions	**Lack of training and digital competence**	Technicians and warehouse staff lack training in digital tools.	"People don't always know how to use the tools we already have."	SQ3	AI Readiness Models
Organizational & Sociotechnical Conditions	**Governance gaps**	No unified governance for data, workflows, or sustainability.	"Governance exists on paper, but it's not enforced across teams."	SQ3	AI Readiness Models
Organizational & Sociotechnical Conditions	**Resistance to change**	Staff resist new processes or AI tools due to habit or fear.	"People prefer the old way, they don't trust automated decisions yet."	SQ3	AI Readiness Models

Theme	Code	Definition	Example Quote (Paraphrased)	Linked SQ	Linked Theory
Organizational & Sociotechnical Conditions	**Misaligned KPIs**	Operational KPIs prioritize speed and cost, not reuse or data quality.	"Teams are measured on speed, not documentation quality or reuse."	SQ3	AI Readiness Models
Organizational & Sociotechnical Conditions	**Customer restrictions on AI use**	Some customers prohibit using their data for AI training.	"Certain customers explicitly forbid us from using their data in AI systems."	SQ3	AI Readiness Models

B2.4 Table - Theme 4: Sustainability-Driven Requirements

(SQ1 + SQ3 + Circular Economy Principles)

Theme	Code	Definition	Example Quote (Paraphrased)	Linked SQ	Linked Theory
Sustainability-Driven Requirements	**Reuse and scrap-reduction targets**	Organizations aim to maximize reuse and minimize scrapping.	"We're under pressure to reuse more and scrap less, but the process isn't aligned."	**SQ1 / SQ3**	**Circular Economy Principles**
Sustainability-Driven Requirements	**CO_2 reporting dependencies**	Accurate condition data is required for CO_2 and ESG reporting.	"If a unit is misclassified, our CO_2 reporting becomes inaccurate."	SQ1 / SQ3	Circular Economy Principles
Sustainability-Driven Requirements	**Circular economy incentives**	CE goals encourage reuse, refurbishment, and component harvesting.	"Refurbished units save around 80% of CO_2 compared to new production."	SQ3	Circular Economy Principles
Sustainability-Driven Requirements	**Conflicts between speed and sustainability**	Operational urgency often overrides	"When timelines are tight, careful dismantling and	SQ3	Circular Economy Principles

Theme	Code	Definition	Example Quote (Paraphrased)	Linked SQ	Linked Theory
		sustainability practices.	documentation are ignored."		
Sustainability-Driven Requirements	**Lack of sustainability KPIs for field teams**	Sustainability goals are not embedded in operational KPIs.	"Sustainability is important, but field teams aren't measured on it."	SQ3	Circular Economy Principles
Sustainability-Driven Requirements	**Component-level reuse opportunities**	Some components can be reused even when full units cannot.	"Even if the radio is faulty, many RF components can still be reused."	SQ1	Circular Economy Principles
Sustainability-Driven Requirements	**Regulatory pressure**	CSRD and EU directives require traceable, auditable data.	"New EU rules mean we must track reuse and CO_2 more accurately."	SQ1 / SQ3	Circular Economy Principles
Sustainability-Driven Requirements	**Misalignment between sustainability and operations**	Sustainability goals conflict with commercial or operational priorities.	"Portfolio teams sometimes block refurbished products to protect new sales."	SQ3	Circular Economy Principles

B.3 Full Narrative Codebook

This section provides detailed descriptions of all 32 codes used in the thematic analysis. Each code includes definition, Inclusion and exclusion criteria, A paraphrased example quote, Theoretical linkage and Sub-question relevance.

Theme 1 - Data-Related Factors

(Linked to SQ1 and Data Quality Framework)

1. Data Fragmentation
Definition: Data relevant to condition assessment is distributed across multiple disconnected systems (field apps, warehouse tools, vendor portals, spreadsheets), preventing a unified view of equipment history.
Inclusion criteria: Mentions of multiple systems not integrated, References to "data islands" or "no single source of truth".

Exclusion criteria: Issues caused by human error alone (covered under other codes)
Paraphrased quote: "Information from field teams, warehouses, and vendors sits in different systems, so we never see the full picture."

2. Missing Metadata
Definition: Critical contextual information (reason for removal, configuration, software version, accessories) is often absent or incomplete.
Inclusion criteria: Missing fields in return documentation, Lack of contextual data needed for AI training.
Exclusion criteria: Incorrect data entries (covered under serial number errors)
Paraphrased quote: "Units arrive with half the fields empty, no reason code, no context, nothing."

3. Inconsistent Defect Labels
Definition: Technicians use free-text descriptions without a standardized taxonomy, leading to inconsistent defect terminology.
Inclusion criteria: Free-text defect descriptions, Multiple labels for the same issue
Exclusion criteria: Missing labels entirely (covered under missing metadata)
Paraphrased quote: "The same defect can be described in ten different ways depending on who typed it."
4. Serial Number Errors
Definition: Manual entry of serial numbers leads to incorrect, unreadable, or mismatched identifiers.
Inclusion criteria: Typing errors, Faded or missing labels requiring manual entry.
Exclusion criteria: Missing metadata unrelated to serials
Paraphrased quote: "We constantly receive units with wrong or unreadable serials because they were typed manually."

5. Poor Photo Documentation
Definition: Photos vary in angle, lighting, and completeness, limiting their usefulness for assessment or AI training.
Inclusion criteria: Blurry or incomplete photos, Lack of standardized photo capture
Exclusion criteria: Missing photos entirely (covered under missing metadata)
Paraphrased quote: "Photos are often blurry or don't show the damaged area clearly."

6. NFF (no fault found) -Related Data Gaps
Definition: Insufficient failure documentation leads to high No-Fault-Found rates during testing.
Inclusion criteria: Cases where units marked faulty pass testing, Lack of structured failure data.
Exclusion criteria: NFF caused by testing limitations (not data-related)
Paraphrased quote: "Many units marked faulty pass testing because the original fault wasn't documented properly."

7. Install-Base Visibility Issues
Definition: Operators and vendors lack accurate, real-time visibility into installed equipment, affecting assessment and planning.
Inclusion criteria: Uncertainty about what is installed at customer sites, Confidentiality restrictions blocking visibility.
Exclusion criteria: Data fragmentation across internal systems (covered separately)

Paraphrased quote: "We don't always know what's actually installed at customer sites, the data is incomplete."

8. Customer Restrictions on Data Sharing
Definition: Some customers prohibit using their data for AI training, limiting dataset completeness.
Inclusion criteria: Legal or contractual restrictions, Customer-imposed data limitations
Exclusion criteria: Internal governance issues (covered under organizational codes)
Paraphrased quote: "Certain customers forbid us from using their data in AI systems, which blocks model training."

Theme 2 - Workflow & Process Characteristics

(Linked to SQ2 and STS Perspective)

1. High Variability in Dismantling Practices
Definition: Field teams dismantle equipment differently across regions, subcontractors, and customer programs.
Inclusion criteria: Differences in dismantling quality, Variability in handling practices
Exclusion criteria: Variability in data (covered under Theme 1)
Paraphrased quote: "How equipment is removed depends entirely on the subcontractor and the site, no two teams do it the same way."

2. Subjective Visual Inspection
Definition: Damage assessment relies heavily on technician judgement with no objective criteria.
Inclusion criteria: Disagreements between inspectors, Ambiguous visual assessments
Exclusion criteria: Tacit knowledge (covered separately)
Paraphrased quote: "Two inspectors can look at the same radio and reach different conclusions about whether it's damaged."

3. Tacit Knowledge Dependence
Definition: Assessment decisions rely on experience, intuition, and undocumented routines.
Inclusion criteria: "Experience-based" decisions, Reliance on technician intuition.
Exclusion criteria: Subjective visual inspection (covered separately)
Paraphrased quote: "A lot of decisions come down to what the technician 'feels' is wrong with the unit."

4. Inconsistent Photo Capture
Definition: Photos are taken differently at each site, with no standard angles or requirements.
Inclusion criteria: Variability in photo quality, Lack of standardized photo guidelines
Exclusion criteria: Poor photo quality alone (Theme 1)
Paraphrased quote: "Photos are taken differently at every site, some show everything, others show almost nothing."

5. Non-Standardized Workflows
Definition: Inspection steps differ across warehouses, regions, and subcontractors.
Inclusion criteria: Different testing routines and Different routing decisions
Exclusion criteria: Organizational governance issues (Theme 3)
Paraphrased quote: "Each warehouse has its own way of doing inspections, there's no unified process."

6. Time Pressure at Field Sites
Definition: Technicians prioritize speed over documentation quality due to operational urgency.
Inclusion criteria: Rushed dismantling, Skipped documentation steps.
Exclusion criteria: Lack of training (Theme 3)
Paraphrased quote: "When a site is behind schedule, documentation becomes the first thing to suffer."

7. Lack of Reference Images
Definition: Inspectors lack standardized reference images or golden samples for comparison.
Inclusion criteria: Absence of visual benchmarks Reliance on memory
Exclusion criteria: Poor photo capture (Theme 1)
Paraphrased quote: "Inspectors rely on memory because we don't have reference images to compare against."

8. ASP–LSP Disputes
Definition: Disagreements occur about when and where damage happened due to lack of traceability.
Inclusion criteria: Disputes between field teams and logistics, Conflicting claims about damage timing.
Exclusion criteria: Data-related disputes (Theme 1)
Paraphrased quote: "ASP says the unit was fine at handover, LSP says it arrived damaged, we have no way to verify."

Theme 3 - Organizational & Sociotechnical Conditions

(Linked to SQ3 and AI Readiness Models)

1. Fragmented Ownership of Return Process

Definition: No single function owns the end-to-end return and assessment workflow.
Inclusion criteria: Multiple teams involved with unclear ownership, Lack of end-to-end accountability.
Exclusion criteria: System fragmentation (Theme 1)
Paraphrased quote: "Field, logistics, warehouse, procurement, everyone owns a piece, but no one owns the whole process."

2. Siloed Systems and Data
Definition: Field, warehouse, and vendor systems are disconnected, limiting data flow.
Inclusion criteria: Lack of integration, Data stored in separate platforms.
Exclusion criteria: Fragmented ownership (covered separately)
Paraphrased quote: "Our systems don't talk to each other, data sits in separate islands."

3. Limited AI Maturity
Definition: Organizations lack AI governance, skills, and infrastructure.
Inclusion criteria: Early-stage AI initiatives, Lack of AI governance frameworks
Exclusion criteria: Lack of training (covered separately)
Paraphrased quote: "We see the potential of AI, but we're not ready to support it operationally."

4. Lack of Training and Digital Competence
Definition: Technicians and warehouse staff lack training in digital tools and structured documentation.
Inclusion criteria: Low digital literacy, Difficulty using existing tools.
Exclusion criteria: Resistance to change (covered separately)
Paraphrased quote: "People don't always know how to use the tools we already have."

5. Governance Gaps
Definition: No unified governance for data, workflows, or sustainability.
Inclusion criteria: Weak enforcement of standards, Lack of cross-functional governance
Exclusion criteria: Misaligned KPIs (covered separately)
Paraphrased quote: "Governance exists on paper, but it's not enforced across teams."

6. Resistance to Change
Definition: Staff resist new processes or AI tools due to habit, fear, or uncertainty.
Inclusion criteria: Preference for manual routines, Distrust of automation
Exclusion criteria: Lack of training (covered separately)
Paraphrased quote: "People prefer the old way, they don't trust automated decisions yet."

7. Misaligned KPIs
Definition: Operational KPIs prioritize speed and cost, not reuse or data quality.
Inclusion criteria: KPIs that conflict with sustainability, KPIs that discourage documentation.
Exclusion criteria: Lack of sustainability KPIs (Theme 4)
Paraphrased quote: "Teams are measured on speed, not documentation quality or reuse."

8. Customer Restrictions on AI Use
Definition: Some customers prohibit using their data for AI training.
Inclusion criteria: Legal restrictions, Contractual limitations
Exclusion criteria: Internal governance issues (covered separately)
Paraphrased quote: "Certain customers explicitly forbid us from using their data in AI systems."

Theme 4 - Sustainability-Driven Requirements

(Linked to SQ1, SQ3, and Circular Economy Principles)

1. Reuse and Scrap-Reduction Targets
Definition: Organizations aim to maximize reuse and minimize scrapping to meet sustainability goals.
Inclusion criteria: Reuse targets, Scrap reduction initiatives
Exclusion criteria: Component-level reuse (covered separately)
Paraphrased quote: "We're under pressure to reuse more and scrap less, but the process isn't aligned."

2. CO_2 Reporting Dependencies
Definition: Accurate condition data is required for CO_2 and ESG reporting.
Inclusion criteria: Scope 3 emissions, Avoided emissions calculations.
Exclusion criteria: General sustainability goals (covered separately)
Paraphrased quote: "If a unit is misclassified, our CO_2 reporting becomes inaccurate."

3. Circular Economy Incentives
Definition: CE goals encourage reuse, refurbishment, and component harvesting.
Inclusion criteria: Refurbishment benefits, CO_2 savings from reuse
Exclusion criteria: Regulatory pressure (covered separately)
Paraphrased quote: "Refurbished units save around 80% of CO_2 compared to new production."

4. Conflicts Between Speed and Sustainability
Definition: Operational urgency often overrides sustainability practices.
Inclusion criteria: Time pressure reducing documentation, Speed prioritized over careful handling.
Exclusion criteria: Lack of sustainability KPIs (covered separately)
Paraphrased quote: "When timelines are tight, careful dismantling and documentation are ignored."

5. Lack of Sustainability KPIs for Field Teams
Definition: Sustainability goals are not embedded in operational KPIs.
Inclusion criteria: No reuse KPIs for technicians, No CO_2-related KPIs for warehouses
Exclusion criteria: Misaligned KPIs in general (Theme 3)
Paraphrased quote: "Sustainability is important, but field teams aren't measured on it."

6. Component-Level Reuse Opportunities
Definition: Some components can be reused even when full units cannot.
Inclusion criteria: RF component reuse, Partial refurbishment
Exclusion criteria: Full-unit reuse (covered separately)
Paraphrased quote: "Even if the radio is faulty, many RF components can still be reused."

7. Regulatory Pressure (CSRD, EU Directives)
Definition: CSRD (Corporate Sustainability Reporting Directive) and EU directives require traceable, auditable data for sustainability reporting.
Inclusion criteria: CSRD, Digital Product Passport EU circularity requirements
Exclusion criteria: Internal sustainability goals (covered separately)
Paraphrased quote: "New EU rules mean we must track reuse and CO_2 more accurately."

8. Misalignment Between Sustainability and Operations
Definition: Sustainability goals conflict with commercial or operational priorities.
Inclusion criteria: Portfolio teams blocking refurbished products, Conflicts between sales and sustainability.
Exclusion criteria: Lack of sustainability KPIs (covered separately)
Paraphrased quote: "Portfolio teams sometimes block refurbished products to protect new sales."